

Return-Adjusted Hierarchical Risk Parity and Schur Portfolios: Mathematical Foundations and Empirical Evidence

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Abstract

This paper develops a fixed-tree theory for Return-Adjusted Hierarchical Risk Parity (RA-HRP), a minimal modification of Lopez de Prado’s [Lopez de Prado, 2016] Hierarchical Risk Parity that preserves clustering and seriation while replacing inverse-variance recursive-bisection scores with floored cluster Sharpe scores. For a fixed dendrogram we prove existence, uniqueness, budget feasibility, and quantitative lower bounds on leaf weights; establish monotonicity and curvature properties of the local split rule; and show that the global weight distortion relative to HRP factorises exactly along root-to-leaf paths, with a companion Kullback–Leibler decomposition into node-level binary divergences. We also develop a Schur-complement extension that incorporates cross-cluster covariance and a spectral companion benchmark framework. Under i.i.d. sampling with finite fourth moments, fixed dimension, and an inactive population floor, the sample fixed-tree RA-HRP estimator is consistent and asymptotically normal at the $\sqrt{T}$ rate. Because the empirical implementation re-estimates the dendrogram each month, we complement the fixed-tree theory with rolling dendrogram-stability diagnostics. In a 15-stock US equity backtest (2015–2024 out of sample) RA-HRP attains a Sharpe ratio of 1.349 versus 1.063 for traditional HRP, but a capped within-universe market-cap proxy achieves a higher Sharpe of 1.571. We therefore interpret the empirical evidence as proof of concept: local return tilts can materially improve HRP on this sample, but the edge is regime-dependent and not a claim of universal dominance.

Keywords: Return-Adjusted Hierarchical Risk Parity, Portfolio Optimisation, Sharpe Ratio Allocation, Schur Complement, Asymptotic Consistency, Cluster Sharpe Ratio, Risk Management

1 Introduction

Portfolio construction remains a central problem in quantitative finance. The classical mean–variance framework of [Markowitz \[1952\]](#) is elegant and foundational, but its practical implementation is often fragile: estimated efficient portfolios depend sensitively on the sample mean and, through matrix inversion, on the conditioning of the covariance matrix. These issues are especially acute when the number of assets is not small relative to the available sample size [[Michaud, 1989](#), [De Miguel et al., 2009](#)]. Remediation strategies include imposing portfolio constraints [[Jagannathan and Ma, 2003](#)] and shrinking the sample covariance matrix toward a structured target [[Ledoit and Wolf, 2004](#)], but the core instability of global inversion persists.

Hierarchical Risk Parity (HRP), introduced by [Lopez de Prado \[2016\]](#), addresses part of this problem by replacing global covariance inversion with hierarchical clustering and recursive bisection. HRP is attractive because it is numerically stable, naturally diversified, and robust to singular or nearly singular covariance matrices [[Antonov et al., 2024](#)]. Its basic limitation, however, is that the recursive allocation step is purely risk-based and does not incorporate expected-return information.

This paper introduces **Return-Adjusted Hierarchical Risk Parity (RA-HRP)**, which modifies only the local allocation step of HRP. The clustering and leaf-ordering stages remain exactly as in HRP, but each sibling-cluster split is determined by cluster-level Sharpe scores rather than inverse variances. The construction is deliberately minimal: it retains the structural stability of HRP while allowing expected return information to influence the final allocation in a transparent and economically interpretable way.

1.1 Contributions

The main contributions of this paper are:

- (i) A rigorous formulation of RA-HRP for a fixed dendrogram on a complete probability space, with proofs of existence, uniqueness, and full-budget feasibility including explicit quantitative lower bounds on leaf weights (Sections 3–5).
- (ii) A full curvature analysis of the local split rule, establishing strict monotonicity, concavity in own score, convexity in sibling score, and a sensitivity bound tied to the floor parameter (Theorem 5.8).
- (iii) A unit-free local interpolation family that nests HRP and RA-HRP as endpoint cases and makes the transition from risk-only to return-adjusted splitting explicit (Proposition 6.2).
- (iv) A pathwise propagation theorem establishing that the global weight ratio $w_i^{\text{RA}}/w_i^{\text{HRP}}$ factorises exactly as a multiplicative product of local split ratios along each root-to-leaf path, together with a KL-divergence decomposition into node-level binary divergences and an explicit necessary and sufficient coincidence condition expressed via cluster Sharpe-to-inverse-variance ratios (Theorem 6.5, Theorem 6.8, and Corollary 6.7).

- (v) A Schur-complement extension that accounts for cross-cluster covariance through conditional risk, with quantitative variance reduction bounds (Section 7).
- (vi) A fixed-tree asymptotic theory under i.i.d. sampling with finite fourth moments, delivering consistency and asymptotic normality at the $\sqrt{T}$ rate with explicit Jacobian structure exploiting the tree recursion, together with Lipschitz stability with an explicit constant (Section 8).
- (vii) Empirical dendrogram stability diagnostics across 119 consecutive rolling windows, assessing the practical relevance of the fixed-tree assumption (Section 11.5).
- (viii) A spectral companion benchmark framework clarifying the relation between decomposition-based benchmark directions and the classical minimum-variance and mean-variance portfolios, together with a simplex projection error bound quantifying how the long-only constraint distorts the unconstrained tangency portfolio (Section 10).
- (ix) An empirical study using 15 major US stocks over 2010–2024, showing that RA-HRP outperforms traditional HRP in the reported backtest while acknowledging that the capped within-universe market-cap proxy achieves a higher Sharpe in the same window (Section 12).

1.2 Paper Structure

Section 2 reviews the related literature. Section 3 introduces notation. Section 4 formalises HRP. Section 5 defines RA-HRP and proves its core properties. Section 6 develops the interpolation family. Section 7 presents the Schur-complement extension. Section 8 studies asymptotics under a fixed dendrogram. Section 9 compares RA-HRP with classical allocation rules. Section 10 introduces the spectral companion benchmarks used in the empirical comparison. Section 11 describes data and methodology. Section 12 presents empirical results. Section 13 discusses economic interpretation and practical implications. Section 14 concludes.

2 Literature Review

Modern portfolio optimisation faces a fundamental dilemma that has persisted since Markowitz [1952]. Methods that incorporate expected returns tend to be highly unstable and prone to extreme concentrations, while methods that achieve stability often ignore return information entirely. Michaud [1989] coined the term “Markowitz optimisation enigma” to describe how optimal portfolios frequently underperform naive alternatives due to estimation errors. De Miguel et al. [2009] provided comprehensive evidence that sophisticated optimisation techniques often fail to outperform simple $1/N$ diversification out-of-sample.

These findings motivated research into robust portfolio construction. Jagannathan and Ma [2003] showed that imposing constraints implicitly addresses estimation errors. Ledoit and Wolf [2004] developed shrinkage estimators for covariance matrices, while Fan et al. [2008] explored factor model approaches. However, the fundamental trade-off between incorporating returns and maintaining stability persists.

Hierarchical Risk Parity, introduced by [Lopez de Prado \[2016\]](#), represents one extreme of this trade-off: achieving remarkable stability by completely excluding expected returns. Its three-step process—hierarchical clustering, quasi-diagonalisation, and recursive bisection—provides a robust alternative that avoids matrix inversion. Recent work by [Antonov et al. \[2024\]](#) provides formal evidence that HRP is less sensitive to estimation errors than traditional Markowitz optimisation. However, HRP’s complete independence from return estimates represents its primary limitation.

Spectral methods have gained attention for their ability to capture underlying factor structure. [Laloux et al. \[1999\]](#) and [Plerou et al. \[2002\]](#) applied random matrix theory to financial correlations. [Avellaneda and Serur \[2020\]](#) developed Hierarchical PCA for modelling asset correlations. Cotton’s [\[Cotton, 2024\]](#) Schur Complementary Allocation provides an elegant unification of HRP and minimum variance approaches through matrix decomposition, demonstrating that both methods can be viewed through the lens of matrix partitioning and Schur complements.

Our work addresses the gap between these approaches: RA-HRP incorporates expected-return information without the global covariance inversion that makes classical mean–variance optimisation fragile. The Schur-complement construction of Section 7 retains a local hierarchical interpretation, while the spectral companion benchmarks of Section 10 provide decomposition-based global comparators for the empirical study.

3 Notation and Preliminaries

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space supporting all random objects below. We consider n risky assets observed over discrete periods $t = 1, \dots, T$.

Definition 3.1 (Return process). Let $\mathbf{r}_t = (r_{1,t}, \dots, r_{n,t})^\top \in \mathbb{R}^n$ denote the vector of one-period *excess returns* at time t . Throughout Sections 3–7 we assume only that $\mathbb{E}\|\mathbf{r}_t\|_2^2 < \infty$ and define

$$\boldsymbol{\mu} := \mathbb{E}[\mathbf{r}_t] \in \mathbb{R}^n, \quad \boldsymbol{\Sigma} := \text{Cov}(\mathbf{r}_t) \in \mathbb{S}_{++}^n,$$

where $\mathbb{S}_{++}^n$ is the cone of real symmetric positive-definite $n \times n$ matrices.

We write $\boldsymbol{\Sigma} = \mathbf{D}\mathbf{P}\mathbf{D}$, where $\mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_n)$ with $\sigma_i = \sqrt{\Sigma_{ii}}$, and $\mathbf{P} = (\rho_{ij})$ is the correlation matrix.

Definition 3.2 (Long-only portfolio). A *long-only fully invested portfolio* is a vector $\mathbf{w} \in \Delta^{n-1}$, where

$$\Delta^{n-1} := \{\mathbf{w} \in \mathbb{R}^n : \mathbf{w}^\top \mathbf{1} = 1, w_i \geq 0 \ \forall i\}.$$

Its expected excess return, variance, volatility, and Sharpe ratio are

$$\mu_p(\mathbf{w}) := \mathbf{w}^\top \boldsymbol{\mu}, \quad \sigma_p^2(\mathbf{w}) := \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}, \quad \sigma_p(\mathbf{w}) := \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}, \quad S_p(\mathbf{w}) := \frac{\mu_p(\mathbf{w})}{\sigma_p(\mathbf{w})}.$$

Definition 3.3 (Risk contributions). For $\mathbf{w} \in \Delta^{n-1}$ with $\sigma_p(\mathbf{w}) > 0$, the *marginal risk contribution* (MRC) and *total risk contribution* (TRC) of asset i are

$$\text{MRC}_i(\mathbf{w}) := \frac{\partial \sigma_p(\mathbf{w})}{\partial w_i} = \frac{(\boldsymbol{\Sigma} \mathbf{w})_i}{\sigma_p(\mathbf{w})}, \quad \text{TRC}_i(\mathbf{w}) := w_i \text{MRC}_i(\mathbf{w}).$$

By Euler's theorem for positively homogeneous functions, $\sum_{i=1}^n \text{TRC}_i(\mathbf{w}) = \sigma_p(\mathbf{w})$.

We denote the asset universe by $\mathcal{A} := \{1, \dots, n\}$ and write $|\mathcal{C}|$ for the cardinality of a subset $\mathcal{C} \subseteq \mathcal{A}$.

4 The Hierarchical Risk Parity Framework

HRP [Lopez de Prado, 2016] consists of three stages: hierarchical clustering of assets, quasi-diagonal ordering of the leaves, and recursive bisection.

4.1 Phase 1: Hierarchical Clustering

Definition 4.1 (Correlation distance). For assets $i, j \in \mathcal{A}$, define the correlation distance

$$d_{ij} := \sqrt{\frac{1 - \rho_{ij}}{2}}, \quad d_{ij} \in [0, 1].$$

Proposition 4.2 (Metric property). *The map $d : \mathcal{A} \times \mathcal{A} \rightarrow [0, 1]$ defined in Definition 4.1 is a metric.*

Proof. Because $\mathbf{P}$ is a correlation matrix, it is positive semidefinite. Hence there exist unit vectors $u_1, \dots, u_n$ in a Euclidean space such that $\rho_{ij} = u_i^\top u_j$ for all i, j (Gram representation). Then

$$d_{ij}^2 = \frac{1 - u_i^\top u_j}{2} = \frac{\|u_i - u_j\|_2^2}{4},$$

so $d_{ij} = \frac{1}{2}\|u_i - u_j\|_2$. Therefore d is a rescaled Euclidean distance and inherits non-negativity, symmetry, and the triangle inequality. Since $\Sigma \in \mathbb{S}_{++}^n$, the correlation matrix is positive definite, which rules out $\rho_{ij} = 1$ for $i \neq j$; thus $d_{ij} = 0$ if and only if $i = j$. $\square$

Economic interpretation. The distance measures diversification benefit between assets. When $\rho_{ij} = 1$, $d_{ij} = 0$ (no diversification); when $\rho_{ij} = -1$, $d_{ij} = 1$ (maximum diversification).

Single-linkage clustering constructs a binary dendrogram $\mathcal{T} = (\mathcal{V}, \mathcal{E})$ by iteratively merging the pair of clusters with smallest inter-cluster distance $d(\mathcal{C}_a, \mathcal{C}_b) := \min_{i \in \mathcal{C}_a, j \in \mathcal{C}_b} d_{ij}$.

4.2 Phase 2: Quasi-Diagonalisation

Let $\pi : \mathcal{A} \rightarrow \mathcal{A}$ be the leaf-order permutation induced by a depth-first traversal of the dendrogram, and let $\tilde{\Sigma} := \Sigma[\pi, \pi]$ denote the permuted covariance matrix.

Lemma 4.3 (Contiguous-subtree ordering). *Under the depth-first leaf ordering π , the leaves belonging to any cluster $\mathcal{C} \in \mathcal{V}$ appear as a contiguous interval in the permuted ordering.*

Proof. Proceed by induction on the subtree rooted at $\mathcal{C}$. If $\mathcal{C}$ is a leaf, the claim is trivial. If $\mathcal{C}$ is an internal node with children $\mathcal{C}_L$ and $\mathcal{C}_R$, a depth-first traversal lists all leaves of one child consecutively, followed by all leaves of the other child consecutively. By the induction hypothesis, the leaves within each child already form contiguous intervals. Their concatenation is therefore a contiguous interval for $\mathcal{C}$. $\square$

4.3 Phase 3: Recursive Bisection

Definition 4.4 (Inverse-variance portfolio weights). For a non-empty cluster $\mathcal{C} \subseteq \mathcal{A}$, define the *inverse-variance portfolio* (IVP) weights by

$$\tilde{w}_i^{(\mathcal{C})} := \frac{\sigma_i^{-2}}{\sum_{j \in \mathcal{C}} \sigma_j^{-2}}, \quad i \in \mathcal{C},$$

and define the corresponding cluster variance

$$\sigma^2(\mathcal{C}) := \sum_{i \in \mathcal{C}} \sum_{j \in \mathcal{C}} \tilde{w}_i^{(\mathcal{C})} \tilde{w}_j^{(\mathcal{C})} \Sigma_{ij}.$$

At each internal node of the dendrogram with child clusters $\mathcal{C}_L$ and $\mathcal{C}_R$, HRP allocates the parent budget $\alpha > 0$ according to

$$\alpha_L^{\text{HRP}} := \alpha \frac{\sigma^{-2}(\mathcal{C}_L)}{\sigma^{-2}(\mathcal{C}_L) + \sigma^{-2}(\mathcal{C}_R)}, \quad \alpha_R^{\text{HRP}} := \alpha - \alpha_L^{\text{HRP}}. \quad (1)$$

The final HRP weight of asset i is the product of the split fractions encountered on the root-to-leaf path leading to i :

$$w_i^{\text{HRP}} = \prod_{k=1}^{K_i} \alpha_{k,i}. \quad (2)$$

5 Return-Adjusted Hierarchical Risk Parity

RA-HRP keeps the clustering and leaf-ordering stages of HRP unchanged and replaces the local score in the recursive bisection step.

5.1 Definition

Definition 5.1 (Cluster expected return and cluster Sharpe score). For a cluster $\mathcal{C}$, define its IVP-weighted expected return by

$$\mu(\mathcal{C}) := \sum_{i \in \mathcal{C}} \tilde{w}_i^{(\mathcal{C})} \mu_i,$$

and its *cluster Sharpe score* by

$$S(\mathcal{C}) := \frac{\mu(\mathcal{C})}{\sigma(\mathcal{C})}.$$

Remark 5.2. Using the IVP weights from Definition 4.4 in both $\mu(\mathcal{C})$ and $\sigma(\mathcal{C})$ ensures internal consistency: $S(\mathcal{C})$ is exactly the Sharpe ratio of the IVP restricted to cluster $\mathcal{C}$.

Definition 5.3 (RA-HRP bisection rule). Fix a score floor $\varepsilon > 0$ and define $S^+(\mathcal{C}) := \max\{S(\mathcal{C}), \varepsilon\}$. At an internal node with child clusters $\mathcal{C}_L$ and $\mathcal{C}_R$, the *RA-HRP* split is

$$\alpha_L^{\text{RA}} := \alpha \frac{S^+(\mathcal{C}_L)}{S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R)}, \quad \alpha_R^{\text{RA}} := \alpha - \alpha_L^{\text{RA}}. \quad (3)$$

Applying (3) recursively at every internal node defines the final weight vector $\mathbf{w}^{\text{RA}} \in \Delta^{n-1}$.

Remark 5.4. The score floor ε guarantees a well-defined positive split even when a raw cluster Sharpe ratio is small or negative. In applications where all population cluster Sharpe ratios are positive and bounded away from zero, the floor is inactive asymptotically but remains useful numerically.

5.2 Core Properties

Theorem 5.5 (Existence and uniqueness). *Let $\Sigma \in \mathbb{S}_{++}^n$, fix a dendrogram $\mathcal{T}$ with depth D , and fix $\varepsilon > 0$. Then RA-HRP defines a unique long-only fully invested portfolio $\mathbf{w}^{\text{RA}} \in \Delta^{n-1}$. Moreover, the weight of every leaf satisfies the quantitative lower bound*

$$w_i^{\text{RA}} \geq \left(\frac{\varepsilon}{\varepsilon + M_S} \right)^D, \quad (4)$$

where $M_S := \max_{\mathcal{C}} S^+(\mathcal{C})$ ranges over all clusters in $\mathcal{T}$ and D is the depth of the deepest leaf.

Proof. **Step 1 (Well-definedness of local splits).** At every internal node, the two floored scores satisfy $S^+(\mathcal{C}_L) \geq \varepsilon$ and $S^+(\mathcal{C}_R) \geq \varepsilon$. Hence the denominator in (3) satisfies $S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R) \geq 2\varepsilon > 0$, and each split fraction lies in the open interval

$$\frac{\varepsilon}{\varepsilon + M_S} \leq \frac{S^+(\mathcal{C}_L)}{S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R)} \leq \frac{M_S}{\varepsilon + M_S} < 1.$$

Step 2 (Recursive termination and positivity). The dendrogram has n leaves and $n - 1$ internal nodes. The recursive procedure starts with root budget $\alpha = 1$ and applies the split (3) at each internal node. Because every split fraction is strictly positive and strictly less than one, each child receives a strictly positive budget that is strictly less than the parent budget. After finitely many steps (at most $n - 1$ splits), a unique positive budget arrives at each leaf.

Step 3 (Quantitative lower bound). Along any root-to-leaf path of length $K_i \leq D$, the budget assigned to leaf i is a product of K_i split fractions. The worst case at each node is the minimum fraction $\varepsilon/(\varepsilon + M_S)$. Therefore

$$w_i^{\text{RA}} = \prod_{k=1}^{K_i} \beta_{k,i}^{\text{RA}} \geq \left(\frac{\varepsilon}{\varepsilon + M_S} \right)^{K_i} \geq \left(\frac{\varepsilon}{\varepsilon + M_S} \right)^D.$$

Step 4 (Uniqueness). For fixed $(\boldsymbol{\mu}, \Sigma, \mathcal{T}, \varepsilon)$, the quantities $\tilde{w}_i^{(\mathcal{C})}$, $\sigma(\mathcal{C})$, $\mu(\mathcal{C})$, and $S^+(\mathcal{C})$ are all deterministic functions of the data. Because the split rule (3) is a deterministic function of these quantities, the recursive construction produces a unique output. $\square$

Theorem 5.6 (Budget feasibility). *Under the hypotheses of Theorem 5.5, $\mathbf{w}^{\text{RA}} \in \Delta^{n-1}$. More precisely, for every subtree rooted at a node v with incoming budget $\alpha_v > 0$, the sum of leaf weights in that subtree equals α_v .*

Proof. We prove by strong induction on the number of leaves m in a subtree that the subtree allocates its incoming budget exactly.

Base case ($m = 1$). A leaf node v receives budget α_v and assigns weight $w_v = \alpha_v$. The sum of leaf weights is α_v .

Inductive step. Suppose the claim holds for all subtrees with fewer than m leaves. Let v be an internal node with incoming budget $\alpha_v > 0$ and children $\mathcal{C}_L, \mathcal{C}_R$ having m_L and m_R leaves respectively, with $m_L + m_R = m$ and $m_L, m_R \geq 1$. The split (3) gives

$$\alpha_L^{\text{RA}} + \alpha_R^{\text{RA}} = \alpha_v \cdot \frac{S^+(\mathcal{C}_L)}{S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R)} + \alpha_v \cdot \frac{S^+(\mathcal{C}_R)}{S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R)} = \alpha_v.$$

By Theorem 5.5, $\alpha_L^{\text{RA}}, \alpha_R^{\text{RA}} > 0$. Since $m_L < m$ and $m_R < m$, the induction hypothesis gives

$$\sum_{i \in \text{leaves}(\mathcal{C}_L)} w_i = \alpha_L^{\text{RA}}, \quad \sum_{i \in \text{leaves}(\mathcal{C}_R)} w_i = \alpha_R^{\text{RA}}.$$

Adding yields $\sum_{i \in \text{leaves}(v)} w_i = \alpha_v$. Setting $\alpha_v = 1$ at the root gives $\mathbf{1}^\top \mathbf{w}^{\text{RA}} = 1$. Combined with the strict positivity from (4), $\mathbf{w}^{\text{RA}} \in \Delta^{n-1}$. $\square$

Proposition 5.7 (Risk contribution decomposition). *For any $\mathbf{w} \in \Delta^{n-1}$ with $\sigma_p(\mathbf{w}) > 0$,*

$$\text{TRC}_i(\mathbf{w}) = \frac{w_i(\Sigma \mathbf{w})_i}{\sigma_p(\mathbf{w})}, \quad \sum_{i=1}^n \text{TRC}_i(\mathbf{w}) = \sigma_p(\mathbf{w}).$$

In particular, the identity holds for $\mathbf{w}^{\text{RA}}$.

Proof. The map $\sigma_p(\mathbf{w}) = (\mathbf{w}^\top \Sigma \mathbf{w})^{1/2}$ is positively homogeneous of degree one. Euler's theorem yields $\mathbf{w}^\top \nabla \sigma_p(\mathbf{w}) = \sigma_p(\mathbf{w})$, which is precisely the stated summation identity. $\square$

Theorem 5.8 (Monotonicity and curvature in sibling scores). *Fix a node with parent budget $\alpha > 0$ and define $a := S^+(\mathcal{C}_L)$ and $b := S^+(\mathcal{C}_R)$. Then the left-cluster allocation share*

$$f(a, b) := \frac{\alpha_L^{\text{RA}}}{\alpha} = \frac{a}{a + b}$$

satisfies the following properties:

- (a) Strict monotonicity: $\partial f / \partial a > 0$ and $\partial f / \partial b < 0$ on $(0, \infty)^2$.
- (b) Concavity in own score: $\partial^2 f / \partial a^2 < 0$, so the marginal allocation gain from increasing a is diminishing.
- (c) Convexity in sibling score: $\partial^2 f / \partial b^2 > 0$, so the marginal allocation loss from increasing b is also diminishing.
- (d) Cross-effect: $\partial^2 f / \partial a \partial b = (a - b) / (a + b)^3$, which changes sign at $a = b$.
- (e) Sensitivity bound: For $a, b \geq \varepsilon$,

$$\left| \frac{\partial f}{\partial a} \right| \leq \frac{1}{4\varepsilon}, \quad \left| \frac{\partial f}{\partial b} \right| \leq \frac{1}{4\varepsilon}.$$

Proof. **Part (a).** Direct differentiation:

$$\frac{\partial f}{\partial a} = \frac{b}{(a+b)^2} > 0, \quad \frac{\partial f}{\partial b} = -\frac{a}{(a+b)^2} < 0,$$

because $a, b \geq \varepsilon > 0$.

Part (b). The second derivative is

$$\frac{\partial^2 f}{\partial a^2} = -\frac{2b}{(a+b)^3} < 0.$$

Part (c). Similarly,

$$\frac{\partial^2 f}{\partial b^2} = \frac{2a}{(a+b)^3} > 0.$$

Part (d). The cross derivative is

$$\frac{\partial^2 f}{\partial a \partial b} = \frac{(a+b)^2 - 2b(a+b)}{(a+b)^4} = \frac{a-b}{(a+b)^3}.$$

Part (e). Since $b/(a+b)^2 \leq 1/(4b)$ by the AM–GM inequality applied to $a+b \geq 2\sqrt{ab}$, we get $b/(a+b)^2 \leq b/(2\sqrt{ab})^2 = 1/(4a)$. More directly, the map $a \mapsto b/(a+b)^2$ is maximised at $a = b$ where it equals $1/(4b) \leq 1/(4\varepsilon)$. The bound on $|\partial f/\partial b|$ follows by symmetry in the roles of a and b . $\square$

Corollary 5.9 (Dominance condition). *If $S^+(\mathcal{C}_L) > S^+(\mathcal{C}_R)$, then $\alpha_L^{\text{RA}} > \alpha/2 > \alpha_R^{\text{RA}}$.*

6 The Interpolation Family

At each internal node with children $(\mathcal{C}_L, \mathcal{C}_R)$, define the dimensionless HRP and RA-HRP left-split fractions

$$r_L := \frac{\sigma^{-2}(\mathcal{C}_L)}{\sigma^{-2}(\mathcal{C}_L) + \sigma^{-2}(\mathcal{C}_R)}, \quad s_L := \frac{S^+(\mathcal{C}_L)}{S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R)}.$$

Both belong to $(0, 1)$, so they can be blended without mixing unlike units.

Definition 6.1 (Unit-free local interpolation). For $\lambda \in [0, 1]$, define

$$\alpha_L^{(\lambda)} := \alpha \left[(1-\lambda)r_L + \lambda s_L \right], \quad \alpha_R^{(\lambda)} := \alpha - \alpha_L^{(\lambda)}.$$

Proposition 6.2 (Unit-free HRP–RA-HRP interpolation). *For the family in Definition 6.1:*

- (i) when $\lambda = 0$, $\alpha_L^{(\lambda)}$ reduces to the HRP split (1);
- (ii) when $\lambda = 1$, $\alpha_L^{(\lambda)}$ reduces to the RA-HRP split (3);
- (iii) the map $\lambda \mapsto \alpha_L^{(\lambda)}$ is affine and therefore continuous on $[0, 1]$; and

(iv) its derivative is

$$\frac{d}{d\lambda} \frac{\alpha_L^{(\lambda)}}{\alpha} = s_L - r_L,$$

so the path is monotone whenever the HRP and RA-HRP split fractions differ.

Proof. Items (i) and (ii) follow from direct substitution. Item (iii) is immediate because $\alpha_L^{(\lambda)}$ is affine in λ , and item (iv) follows by differentiation. $\square$

Remark 6.3 (Interpretation). Because the interpolation is applied to the split fractions rather than to raw score magnitudes, λ has a literal meaning: the local convex weight placed on the return-adjusted split relative to the risk-only split. This avoids assigning economic meaning to a blend of quantities with different units such as inverse variances and Sharpe ratios.

Remark 6.4 (Auxiliary score-space homotopy). For numerical experiments one may also consider the algebraic homotopy

$$\phi_\lambda(\mathcal{C}) := (1 - \lambda)\sigma^{-2}(\mathcal{C}) + \lambda S^+(\mathcal{C}),$$

followed by the induced split

$$\tilde{\alpha}_L^{(\lambda)} := \alpha \frac{\phi_\lambda(\mathcal{C}_L)}{\phi_\lambda(\mathcal{C}_L) + \phi_\lambda(\mathcal{C}_R)}.$$

It has the same endpoint portfolios but blends objects with different units, so we treat it only as an implementation-specific diagnostic device rather than as the canonical interpolation.

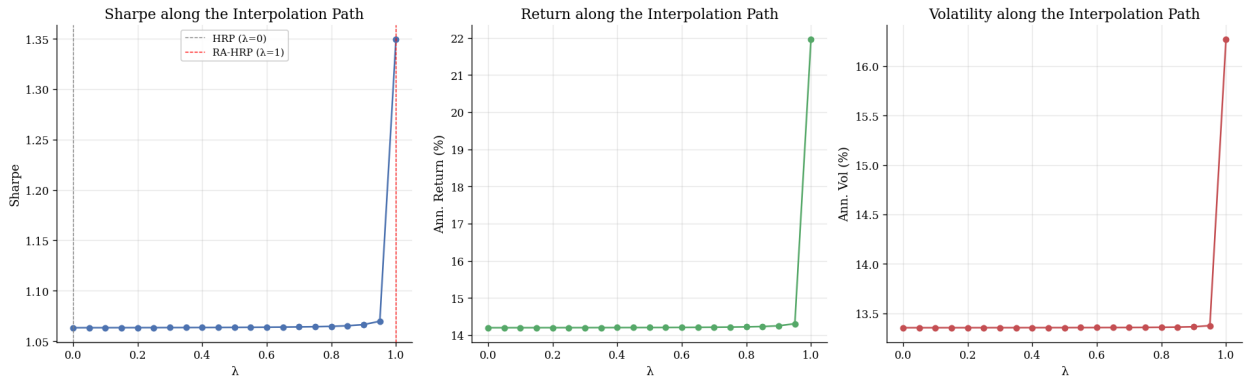


Figure 1: Illustrative HRP–RA-HRP homotopy between the two endpoint portfolios on the 15-stock sample. Left: out-of-sample Sharpe ratio as a function of $\lambda \in [0, 1]$. Centre: annualised return. Right: annualised volatility. The exact shape of the path depends on the normalization used in implementation, so the figure should be read as diagnostic rather than as a claim that every admissible interpolation is monotone in performance.

Theorem 6.5 (Pathwise propagation of local split changes). *Fix a dendrogram $\mathcal{T}$ and consider two hierarchical allocation rules A and B defined on the same tree, with local left-branch*

split fractions $\beta_v^A, \beta_v^B \in (0, 1)$ at each internal node v . For any leaf i , let $\mathcal{P}(i)$ denote the internal nodes on the root-to-leaf path leading to i , and define

$$\eta_v(i) := \begin{cases} 1, & \text{if leaf } i \text{ descends from the left child of node } v, \\ 0, & \text{if leaf } i \text{ descends from the right child of node } v. \end{cases}$$

Then:

(a) Multiplicative factorisation. The final leaf weights satisfy

$$\frac{w_i^A}{w_i^B} = \prod_{v \in \mathcal{P}(i)} \left(\frac{\beta_v^A}{\beta_v^B} \right)^{\eta_v(i)} \left(\frac{1 - \beta_v^A}{1 - \beta_v^B} \right)^{1 - \eta_v(i)}. \quad (5)$$

(b) Additive log-decomposition. Comparing RA-HRP with HRP on the same tree,

$$\log \frac{w_i^{\text{RA}}}{w_i^{\text{HRP}}} = \sum_{v \in \mathcal{P}(i)} \eta_v(i) \log \frac{\beta_v^{\text{RA}}}{\beta_v^{\text{HRP}}} + (1 - \eta_v(i)) \log \frac{1 - \beta_v^{\text{RA}}}{1 - \beta_v^{\text{HRP}}}. \quad (6)$$

(c) Global divergence decomposition. The Kullback–Leibler divergence between the two allocation vectors, viewed as probability distributions on the leaves, decomposes as

$$D_{\text{KL}}(\mathbf{w}^A \parallel \mathbf{w}^B) = \sum_{v \in \mathcal{V}_{\text{int}}} W_v^A \cdot d_{\text{KL}}(\beta_v^A \parallel \beta_v^B), \quad (7)$$

where $\mathcal{V}_{\text{int}}$ is the set of internal nodes, $W_v^A := \sum_{i: v \in \mathcal{P}(i)} w_i^A$ is the total weight assigned by rule A to the subtree rooted at v , and

$$d_{\text{KL}}(p \parallel q) := p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}$$

is the binary KL divergence between the local split fractions.

Proof. Parts (a) and (b). Under any hierarchical recursion on a fixed binary tree, the weight of a leaf is the product of the branch fractions encountered along its root-to-leaf path:

$$w_i^A = \prod_{v \in \mathcal{P}(i)} \left[\beta_v^A \right]^{\eta_v(i)} \left[1 - \beta_v^A \right]^{1 - \eta_v(i)}.$$

Dividing w_i^A by w_i^B yields (5); taking logarithms gives (6).

Part (c). Write the KL divergence explicitly:

$$D_{\text{KL}}(\mathbf{w}^A \parallel \mathbf{w}^B) = \sum_{i=1}^n w_i^A \log \frac{w_i^A}{w_i^B} = \sum_{i=1}^n w_i^A \sum_{v \in \mathcal{P}(i)} \left[\eta_v(i) \log \frac{\beta_v^A}{\beta_v^B} + (1 - \eta_v(i)) \log \frac{1 - \beta_v^A}{1 - \beta_v^B} \right],$$

where we used part (b) to substitute the log-ratio. Exchange the order of summation:

$$= \sum_{v \in \mathcal{V}_{\text{int}}} \left[\log \frac{\beta_v^A}{\beta_v^B} \sum_{i: v \in \mathcal{P}(i), \eta_v(i)=1} w_i^A + \log \frac{1 - \beta_v^A}{1 - \beta_v^B} \sum_{i: v \in \mathcal{P}(i), \eta_v(i)=0} w_i^A \right].$$

The inner sums are the total weight assigned by A to the left and right subtrees of node v , which are $W_v^A \beta_v^A$ and $W_v^A(1 - \beta_v^A)$ respectively. Substituting and simplifying:

$$= \sum_{v \in \mathcal{V}_{\text{int}}} W_v^A \left[\beta_v^A \log \frac{\beta_v^A}{\beta_v^B} + (1 - \beta_v^A) \log \frac{1 - \beta_v^A}{1 - \beta_v^B} \right] = \sum_{v \in \mathcal{V}_{\text{int}}} W_v^A \cdot d_{\text{KL}}(\beta_v^A \parallel \beta_v^B).$$

□

Remark 6.6 (Interpretation of the KL decomposition). Equation (7) shows that the global distance between two hierarchical portfolios decomposes exactly into a weighted sum of local binary divergences. The weight W_v^A measures the capital passing through node v under rule A , so nodes near the root contribute most to the total divergence. Each binary KL term $d_{\text{KL}}(\beta_v^A \parallel \beta_v^B)$ is non-negative and vanishes if and only if $\beta_v^A = \beta_v^B$. This yields a *strictly additive* attribution of the total portfolio distance to individual nodes in the hierarchy.

Corollary 6.7 (Coincidence condition for HRP and RA-HRP). *Suppose that for every internal node with children $(\mathcal{C}_L, \mathcal{C}_R)$,*

$$\frac{S^+(\mathcal{C}_L)}{S^+(\mathcal{C}_R)} = \frac{\sigma^{-2}(\mathcal{C}_L)}{\sigma^{-2}(\mathcal{C}_R)}.$$

Then RA-HRP and HRP generate exactly the same final portfolio on that tree.

Proof. The stated ratio equality implies $\beta_v^{\text{RA}} = \beta_v^{\text{HRP}}$ at every internal node v . Theorem 6.5 then gives $w_i^{\text{RA}} = w_i^{\text{HRP}}$ for every leaf i . □

Theorem 6.8 (Global weight-distortion bound from local split perturbations). *Fix a dendrogram $\mathcal{T}$ and an asset i . Let $\mathcal{P}(i)$ denote the set of internal nodes on the root-to-leaf path leading to i . For each $u \in \mathcal{P}(i)$, let $\beta_u^{\text{HRP}} \in (0, 1)$ and $\beta_u^{\text{RA}} \in (0, 1)$ denote the branch fractions assigned by HRP and RA-HRP respectively at node u for the branch containing asset i .*

Then:

(a) Exact factorisation. *The corresponding leaf weights satisfy*

$$\frac{w_i^{\text{RA}}}{w_i^{\text{HRP}}} = \prod_{u \in \mathcal{P}(i)} \frac{\beta_u^{\text{RA}}}{\beta_u^{\text{HRP}}}.$$

(b) Exponential sandwich bound. *If for every $u \in \mathcal{P}(i)$, $\left| \log \frac{\beta_u^{\text{RA}}}{\beta_u^{\text{HRP}}} \right| \leq \eta_u$, then*

$$e^{-\sum_{u \in \mathcal{P}(i)} \eta_u} \leq \frac{w_i^{\text{RA}}}{w_i^{\text{HRP}}} \leq e^{\sum_{u \in \mathcal{P}(i)} \eta_u}.$$

(c) Explicit score-ratio characterisation. *At each internal node u with children $\mathcal{C}_L, \mathcal{C}_R$, suppose asset i descends from $\mathcal{C}_L$. Then the local log-distortion at node u is*

$$\log \frac{\beta_u^{\text{RA}}}{\beta_u^{\text{HRP}}} = \log \frac{S^+(\mathcal{C}_L) [\sigma^{-2}(\mathcal{C}_L) + \sigma^{-2}(\mathcal{C}_R)]}{\sigma^{-2}(\mathcal{C}_L) [S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R)]}. \quad (8)$$

Defining the cluster Sharpe-to-inverse-variance ratio $\rho(\mathcal{C}) := S^+(\mathcal{C})/\sigma^{-2}(\mathcal{C})$ and letting $\omega_L^{\text{HRP}} := \sigma^{-2}(\mathcal{C}_L)/[\sigma^{-2}(\mathcal{C}_L) + \sigma^{-2}(\mathcal{C}_R)]$ denote the HRP left-branch fraction, the local distortion simplifies to

$$\log \frac{\beta_u^{\text{RA}}}{\beta_u^{\text{HRP}}} = \log \frac{\rho(\mathcal{C}_L)}{\omega_L^{\text{HRP}} \rho(\mathcal{C}_L) + (1 - \omega_L^{\text{HRP}}) \rho(\mathcal{C}_R)}, \quad (9)$$

which is positive if and only if $\rho(\mathcal{C}_L)$ exceeds the HRP-weighted average of the two ρ -values.

Proof. Part (a). This is immediate from the recursive construction: each leaf weight is the product of branch fractions along its root-to-leaf path, and taking ratios gives a telescoping product.

Part (b). Taking logarithms of part (a) and applying the triangle inequality:

$$\left| \log \frac{w_i^{\text{RA}}}{w_i^{\text{HRP}}} \right| = \left| \sum_{u \in \mathcal{P}(i)} \log \frac{\beta_u^{\text{RA}}}{\beta_u^{\text{HRP}}} \right| \leq \sum_{u \in \mathcal{P}(i)} \eta_u.$$

Exponentiating both sides of $-\sum \eta_u \leq \log(w_i^{\text{RA}}/w_i^{\text{HRP}}) \leq \sum \eta_u$ gives the multiplicative bound.

Part (c). The RA-HRP left-branch fraction is $\beta_u^{\text{RA}} = S^+(\mathcal{C}_L)/[S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R)]$ and the HRP left-branch fraction is $\beta_u^{\text{HRP}} = \sigma^{-2}(\mathcal{C}_L)/[\sigma^{-2}(\mathcal{C}_L) + \sigma^{-2}(\mathcal{C}_R)]$. Taking their ratio yields (8) after straightforward algebra. For (9), divide numerator and denominator of (8) by $\sigma^{-2}(\mathcal{C}_L)$ and substitute the definition $\rho(\mathcal{C}) := S^+(\mathcal{C})/\sigma^{-2}(\mathcal{C})$. The denominator inside the logarithm becomes the HRP-weighted average $\omega_L^{\text{HRP}} \rho(\mathcal{C}_L) + (1 - \omega_L^{\text{HRP}}) \rho(\mathcal{C}_R)$. Since $\log(x/\bar{x}) > 0$ iff $x > \bar{x}$, the sign characterisation follows. $\square$

Remark 6.9 (Interpretation). Part (c) reveals the precise mechanism by which return adjustment propagates: the local distortion at each node depends on whether the left cluster's Sharpe-to-inverse-variance ratio exceeds the HRP-weighted average. The quantity $\rho(\mathcal{C}) = S^+(\mathcal{C}) \cdot \sigma^2(\mathcal{C})$ can be interpreted as the cluster's risk-adjusted return scaled by its variance. When $\rho(\mathcal{C}_L) = \rho(\mathcal{C}_R)$, the distortion vanishes — this recovers the coincidence condition of Corollary 6.7 in a more transparent form.

7 Schur-Complement Extension

The basic RA-HRP split treats sibling clusters through standalone cluster risk. A natural refinement, related to the Schur complementary allocation framework of Cotton [2024], is to replace standalone risk by conditional risk that accounts for cross-cluster covariance.

Definition 7.1 (Schur complement). Partition Σ conformably with sibling clusters $\mathcal{C}_L$ and $\mathcal{C}_R$:

$$\Sigma = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{RL} & \Sigma_{RR} \end{pmatrix}.$$

The Schur complement of Σ_{LL} in Σ is

$$\Sigma/\Sigma_{LL} := \Sigma_{RR} - \Sigma_{RL} \Sigma_{LL}^{-1} \Sigma_{LR}.$$

Theorem 7.2 (Variance decomposition and conditional risk reduction). *For any portfolio split as $\mathbf{w} = (\mathbf{w}_L^\top, \mathbf{w}_R^\top)^\top$,*

$$\sigma_p^2(\mathbf{w}) = \mathbf{w}_L^\top \Sigma_{LL} \mathbf{w}_L + 2\mathbf{w}_L^\top \Sigma_{LR} \mathbf{w}_R + \mathbf{w}_R^\top \Sigma_{RR} \mathbf{w}_R. \quad (10)$$

Moreover, if $\Sigma \in \mathbb{S}_{++}^n$, then $\mathbf{w}_R^\top (\Sigma / \Sigma_{LL}) \mathbf{w}_R \leq \mathbf{w}_R^\top \Sigma_{RR} \mathbf{w}_R$ for every $\mathbf{w}_R$, and similarly after exchanging left and right.

Proof. Equation (10) is the block expansion of $\mathbf{w}^\top \Sigma \mathbf{w}$. Because $\Sigma \in \mathbb{S}_{++}^n$, the principal block Σ_{LL} is positive definite and invertible. The Schur complement identity yields $\Sigma_{RR} - (\Sigma / \Sigma_{LL}) = \Sigma_{RL} \Sigma_{LL}^{-1} \Sigma_{LR}$. The right-hand side is positive semidefinite because for every vector x , $x^\top \Sigma_{RL} \Sigma_{LL}^{-1} \Sigma_{LR} x = (\Sigma_{LR} x)^\top \Sigma_{LL}^{-1} (\Sigma_{LR} x) \geq 0$. Hence $\Sigma / \Sigma_{LL} \preceq \Sigma_{RR}$, and the stated quadratic-form inequality follows. $\square$

Definition 7.3 (Symmetric Schur-RA-HRP). Let $\tilde{\mathbf{w}}_L$ and $\tilde{\mathbf{w}}_R$ denote the IVP weight vectors on the two sibling clusters. Define the conditional cluster volatilities

$$\sigma_{\text{Sch}}^2(\mathcal{C}_L | \mathcal{C}_R) := \tilde{\mathbf{w}}_L^\top (\Sigma / \Sigma_{RR}) \tilde{\mathbf{w}}_L, \quad \sigma_{\text{Sch}}^2(\mathcal{C}_R | \mathcal{C}_L) := \tilde{\mathbf{w}}_R^\top (\Sigma / \Sigma_{LL}) \tilde{\mathbf{w}}_R,$$

the conditional Sharpe scores

$$S_{\text{Sch}}(\mathcal{C}_L | \mathcal{C}_R) := \frac{\mu(\mathcal{C}_L)}{\sigma_{\text{Sch}}(\mathcal{C}_L | \mathcal{C}_R)}, \quad S_{\text{Sch}}(\mathcal{C}_R | \mathcal{C}_L) := \frac{\mu(\mathcal{C}_R)}{\sigma_{\text{Sch}}(\mathcal{C}_R | \mathcal{C}_L)},$$

and the floored scores $S_{\text{Sch}}^+(\cdot) := \max\{S_{\text{Sch}}(\cdot), \varepsilon\}$. The symmetric Schur-RA-HRP split is

$$\alpha_L^{\text{Sch}} := \alpha \frac{S_{\text{Sch}}^+(\mathcal{C}_L | \mathcal{C}_R)}{S_{\text{Sch}}^+(\mathcal{C}_L | \mathcal{C}_R) + S_{\text{Sch}}^+(\mathcal{C}_R | \mathcal{C}_L)}, \quad \alpha_R^{\text{Sch}} := \alpha - \alpha_L^{\text{Sch}}. \quad (11)$$

Corollary 7.4 (Quantitative score improvement under Schur adjustment). *For each sibling cluster, conditional variance is no larger than standalone variance:*

$$\sigma_{\text{Sch}}^2(\mathcal{C}_R | \mathcal{C}_L) = \sigma^2(\mathcal{C}_R) - \tilde{\mathbf{w}}_R^\top \Sigma_{RL} \Sigma_{LL}^{-1} \Sigma_{LR} \tilde{\mathbf{w}}_R \leq \sigma^2(\mathcal{C}_R),$$

with equality if and only if $\Sigma_{LR} \tilde{\mathbf{w}}_R = \mathbf{0}$ (i.e., the IVP of the right cluster is uncorrelated with all assets in the left cluster).

Consequently, when the raw cluster Sharpe ratio is positive, the Schur-adjusted score is weakly larger:

$$S_{\text{Sch}}^+(\mathcal{C}_L | \mathcal{C}_R) \geq S^+(\mathcal{C}_L), \quad S_{\text{Sch}}^+(\mathcal{C}_R | \mathcal{C}_L) \geq S^+(\mathcal{C}_R).$$

The variance reduction admits the bound

$$\sigma^2(\mathcal{C}_R) - \sigma_{\text{Sch}}^2(\mathcal{C}_R | \mathcal{C}_L) \leq \frac{\|\Sigma_{LR} \tilde{\mathbf{w}}_R\|_2^2}{\lambda_{\min}(\Sigma_{LL})},$$

so the Schur adjustment is strongest when cross-cluster covariance is large relative to the conditioning cluster's eigenvalue spread.

Proof. The variance reduction term is $\tilde{\mathbf{w}}_R^\top \Sigma_{RL} \Sigma_{LL}^{-1} \Sigma_{LR} \tilde{\mathbf{w}}_R = \|\Sigma_{LL}^{-1/2} \Sigma_{LR} \tilde{\mathbf{w}}_R\|_2^2 \geq 0$, with equality iff $\Sigma_{LR} \tilde{\mathbf{w}}_R = \mathbf{0}$. This follows from positive definiteness of Σ_{LL} .

For the upper bound, write $\Sigma_{LL}^{-1} \preceq \lambda_{\min}(\Sigma_{LL})^{-1} I$, so

$$\tilde{\mathbf{w}}_R^\top \Sigma_{RL} \Sigma_{LL}^{-1} \Sigma_{LR} \tilde{\mathbf{w}}_R \leq \frac{\|\Sigma_{LR} \tilde{\mathbf{w}}_R\|_2^2}{\lambda_{\min}(\Sigma_{LL})}.$$

For the score improvement, if $S(\mathcal{C}_R) > 0$, then $\mu(\mathcal{C}_R) > 0$ and $S_{\text{Sch}}(\mathcal{C}_R \mid \mathcal{C}_L) = \mu(\mathcal{C}_R)/\sigma_{\text{Sch}}(\mathcal{C}_R \mid \mathcal{C}_L) \geq \mu(\mathcal{C}_R)/\sigma(\mathcal{C}_R) = S(\mathcal{C}_R)$. If $S(\mathcal{C}_R) \leq 0$, the floor leaves it at ε unless the Schur adjustment pushes it above ε , in which case it increases. Hence the floored score never decreases. $\square$

8 Asymptotic Theory

8.1 Regularity Assumptions

Assumption 8.1 (Regular fixed-dimension market). The following conditions hold:

- (a) the number of assets n is fixed;
- (b) the dendrogram $\mathcal{T}$ used by RA-HRP is fixed and does not depend on the sample;
- (c) $\{\mathbf{r}_t\}_{t \geq 1}$ are i.i.d. with $\mathbb{E}\|\mathbf{r}_t\|_2^4 < \infty$;
- (d) there exist constants $0 < c_1 \leq c_2 < \infty$ such that $c_1 \leq \lambda_{\min}(\Sigma) \leq \lambda_{\max}(\Sigma) \leq c_2$;
- (e) there exist constants $0 < \delta \leq M < \infty$ such that $\delta \leq \mu_i \leq M$ for $i = 1, \dots, n$;
- (f) the floor parameter satisfies $0 < \varepsilon < \delta/\sqrt{c_2}$.

Remark 8.2. Assumption 8.1(d) controls conditioning, while (e) guarantees positive expected excess returns. For any cluster $\mathcal{C}$ of size m , the IVP weights lie in the m -simplex and therefore satisfy $\|\tilde{\mathbf{w}}_{\mathcal{C}}\|_2^2 \geq 1/m$. By Cauchy interlacing [Vershynin, 2018, Theorem 4.3.28], $\lambda_{\min}(\Sigma_{\mathcal{C}\mathcal{C}}) \geq c_1$ and $\lambda_{\max}(\Sigma_{\mathcal{C}\mathcal{C}}) \leq c_2$. Hence $\sigma(\mathcal{C})$ is bounded away from zero uniformly over clusters. Moreover, $\mu(\mathcal{C}) \geq \delta$ and therefore $S(\mathcal{C}) \geq \delta/\sqrt{c_2} > \varepsilon$. Thus the floor is inactive at the population parameter and the RA-HRP map is locally smooth around $(\boldsymbol{\mu}, \Sigma)$.

8.2 Consistency and Asymptotic Normality

Theorem 8.3 (Consistency and $\sqrt{T}$ asymptotics). *Let Assumption 8.1 hold. Define the sample estimators*

$$\hat{\boldsymbol{\mu}}_T := \frac{1}{T} \sum_{t=1}^T \mathbf{r}_t, \quad \hat{\Sigma}_T := \frac{1}{T-1} \sum_{t=1}^T (\mathbf{r}_t - \hat{\boldsymbol{\mu}}_T)(\mathbf{r}_t - \hat{\boldsymbol{\mu}}_T)^\top,$$

and let $\hat{\mathbf{w}}_T^{\text{RA}}$ be the RA-HRP allocation computed from $(\hat{\boldsymbol{\mu}}_T, \hat{\Sigma}_T)$ using the same fixed dendrogram $\mathcal{T}$. Then:

(i) **Consistency:** $\hat{\mathbf{w}}_T^{\text{RA}} \xrightarrow{\mathbb{P}} \mathbf{w}^{\text{RA}}$.

(ii) **Rate:** $\|\hat{\mathbf{w}}_T^{\text{RA}} - \mathbf{w}^{\text{RA}}\|_2 = O_{\mathbb{P}}(T^{-1/2})$.

(iii) **Asymptotic normality:**

$$\sqrt{T}(\hat{\mathbf{w}}_T^{\text{RA}} - \mathbf{w}^{\text{RA}}) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V})$$

where $\mathbf{V} = J_{\Phi}(\theta) \Psi J_{\Phi}(\theta)^{\top}$, J_{Φ} is the Jacobian of the RA-HRP map, and Ψ is the asymptotic covariance of $\sqrt{T}(\hat{\theta}_T - \theta)$.

Proof. Let $\theta := (\boldsymbol{\mu}, \text{vech}(\boldsymbol{\Sigma})) \in \mathbb{R}^p$ with $p = n + n(n+1)/2$, $\hat{\theta}_T := (\hat{\boldsymbol{\mu}}_T, \text{vech}(\hat{\boldsymbol{\Sigma}}_T))$, and define the fixed-tree RA-HRP map by $\Phi(\theta) := \mathbf{w}^{\text{RA}} \in \mathbb{R}^n$.

Step 1: Asymptotics of the inputs. Under Assumption 8.1(c), the strong law of large numbers gives $\hat{\theta}_T \xrightarrow{a.s.} \theta$. The finite fourth-moment condition $\mathbb{E}\|\mathbf{r}_t\|^4 < \infty$ ensures that the joint vector $(\mathbf{r}_t, \text{vech}(\mathbf{r}_t \mathbf{r}_t^{\top}))$ has finite second moments, so the multivariate CLT yields $\sqrt{T}(\hat{\theta}_T - \theta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Psi)$, where $\Psi_{ij} = \text{Cov}([\mathbf{r}_t, \text{vech}(\mathbf{r}_t \mathbf{r}_t^{\top})]_i, [\mathbf{r}_t, \text{vech}(\mathbf{r}_t \mathbf{r}_t^{\top})]_j)$ after the standard bias adjustment.

Step 2: Smoothness of the fixed-tree map. The map Φ is a composition of maps $\Phi = \Phi_n \circ \dots \circ \Phi_1$, where each Φ_k corresponds to the split at one internal node. We verify that each component is smooth in a neighbourhood of θ .

At each internal node v with children $\mathcal{C}_L, \mathcal{C}_R$, the local split fraction is $\beta_v = S^+(\mathcal{C}_L)/[S^+(\mathcal{C}_L) + S^+(\mathcal{C}_R)]$. Under Assumption 8.1(e)–(f), every population cluster Sharpe ratio satisfies $S(\mathcal{C}) \geq \delta/\sqrt{c_2} > \varepsilon$, so the floor is inactive at θ . In a sufficiently small neighbourhood of θ , the floor remains inactive by continuity. Hence $S^+(\mathcal{C}) = S(\mathcal{C}) = \mu(\mathcal{C})/\sigma(\mathcal{C})$, which is a C^∞ function of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ because:

- $\mu(\mathcal{C})$ is linear in $\boldsymbol{\mu}$ (with coefficients depending smoothly on $\boldsymbol{\Sigma}$ through the IVP weights);
- $\sigma(\mathcal{C}) > 0$ is a smooth function of $\boldsymbol{\Sigma}$ because the IVP weights $\tilde{w}_i^{(\mathcal{C})} = \sigma_i^{-2}/\sum_j \sigma_j^{-2}$ are smooth when all $\sigma_i > 0$ (guaranteed by $\lambda_{\min}(\boldsymbol{\Sigma}) \geq c_1$);
- the quotient $\mu(\mathcal{C})/\sigma(\mathcal{C})$ is smooth because $\sigma(\mathcal{C}) \geq \sqrt{c_1/n} > 0$.

The split fraction $\beta_v = a/(a+b)$ with $a, b > 0$ is smooth in (a, b) . Therefore Φ is C^1 on a neighbourhood of θ .

Step 3: Jacobian structure. By the chain rule, the Jacobian $J_{\Phi}(\theta)$ factorises through the tree structure. Because the weight of leaf i is $w_i = \prod_{v \in \mathcal{P}(i)} \beta_v^{\eta_v(i)} (1 - \beta_v)^{1 - \eta_v(i)}$, the partial derivative with respect to any parameter θ_j is

$$\frac{\partial w_i}{\partial \theta_j} = w_i \sum_{v \in \mathcal{P}(i)} \frac{\partial \log \beta_v^{\eta_v(i)} (1 - \beta_v)^{1 - \eta_v(i)}}{\partial \theta_j} = w_i \sum_{v \in \mathcal{P}(i)} \left[\frac{\eta_v(i)}{\beta_v} - \frac{1 - \eta_v(i)}{1 - \beta_v} \right] \frac{\partial \beta_v}{\partial \theta_j}.$$

Each $\partial \beta_v / \partial \theta_j$ involves only the cluster-level statistics at node v and is uniformly bounded on compact subsets of the parameter space. Therefore the full Jacobian is bounded: $\|J_{\Phi}(\theta)\|_{\text{op}} < \infty$.

Step 4: Consistency. By Step 2 the map Φ is continuous. Since $\hat{\theta}_T \xrightarrow{a.s.} \theta$, the continuous mapping theorem gives $\hat{\mathbf{w}}_T^{\text{RA}} = \Phi(\hat{\theta}_T) \xrightarrow{a.s.} \Phi(\theta) = \mathbf{w}^{\text{RA}}$.

Step 5: Delta method. Because Φ is differentiable at θ with bounded Jacobian $J_\Phi(\theta)$, the multivariate delta method gives

$$\sqrt{T}(\hat{\mathbf{w}}_T^{\text{RA}} - \mathbf{w}^{\text{RA}}) = J_\Phi(\theta)\sqrt{T}(\hat{\theta}_T - \theta) + o_{\mathbb{P}}(1) \xrightarrow{d} \mathcal{N}(\mathbf{0}, J_\Phi(\theta) \Psi J_\Phi(\theta)^\top).$$

The rate $O_{\mathbb{P}}(T^{-1/2})$ follows because $\|\hat{\mathbf{w}}_T^{\text{RA}} - \mathbf{w}^{\text{RA}}\|_2 \leq \|J_\Phi(\theta)\|_{\text{op}}\|\hat{\theta}_T - \theta\|_2 + o_{\mathbb{P}}(\|\hat{\theta}_T - \theta\|_2)$ and $\|\hat{\theta}_T - \theta\|_2 = O_{\mathbb{P}}(T^{-1/2})$. $\square$

Remark 8.4 (Scope of the asymptotic result). Theorem 8.3 is a theorem about the *fixed-tree* estimator. The monthly backtest in Sections 11–12 re-estimates the dendrogram at each date, so the implemented portfolio map is only piecewise smooth and can jump when the clustering topology changes. The dendrogram-stability diagnostics of Section 11.5 are therefore the empirical bridge between the fixed-tree limit theory and the estimated-tree strategy actually traded.

Remark 8.5 (Economic interpretation). Consistency means RA-HRP converges to a well-defined population portfolio as the estimation window grows. Because $\|\hat{\mathbf{w}}_T^{\text{RA}} - \mathbf{w}^{\text{RA}}\|_2 = O_{\mathbb{P}}(T^{-1/2})$, the portfolio weight revision attributable to sampling error shrinks at the parametric rate. By contrast, for MVO the analogous first-order sensitivity can deteriorate with $\|\Sigma^{-1}\|_{\text{op}}$ and hence with the conditioning of Σ , because the Jacobian of the tangency map contains Σ^{-1} . Asymptotic normality also makes confidence intervals for $\mathbf{w}^{\text{RA}}$ computable via the sandwich formula $J_\Phi(\hat{\theta}_T) \hat{\Psi}_T J_\Phi(\hat{\theta}_T)^\top$, enabling turnover-aware rebalancing rules.

Proposition 8.6 (Local Lipschitz stability with explicit constant). *Under Assumption 8.1, there exist a neighbourhood U of $\theta = (\boldsymbol{\mu}, \text{vech}(\Sigma))$ and a finite constant L such that for all $\theta_1, \theta_2 \in U$,*

$$\|\Phi(\theta_1) - \Phi(\theta_2)\|_2 \leq L \|\theta_1 - \theta_2\|_2.$$

The Lipschitz constant satisfies

$$L \leq n \cdot D \cdot \frac{M_S}{4\varepsilon^2} \cdot \sup_{\vartheta \in U} \max_v \left\| \frac{\partial \beta_v}{\partial \theta} \right\|_2,$$

where D is the tree depth, M_S is the maximal score, and the supremum of the score-gradient norms is finite on U by smoothness.

Proof. By Theorem 8.3, the map Φ is C^1 on a neighbourhood of θ . Choose a compact convex subset U of this neighbourhood.

The Jacobian entry for leaf i and parameter θ_j was computed in the proof of Theorem 8.3:

$$\frac{\partial w_i}{\partial \theta_j} = w_i \sum_{v \in \mathcal{P}(i)} g_v(i) \frac{\partial \beta_v}{\partial \theta_j},$$

where $g_v(i) := \eta_v(i)/\beta_v - (1 - \eta_v(i))/(1 - \beta_v)$ satisfies $|g_v(i)| \leq 1/\min(\beta_v, 1 - \beta_v) \leq (\varepsilon + M_S)/\varepsilon$ by the bounds in the proof of Theorem 5.5. Since $w_i \leq 1$, $|\mathcal{P}(i)| \leq D$, and the gradient norms are bounded on U , the operator norm of J_Φ is bounded uniformly on U . The mean-value theorem on the convex set U then gives $\|\Phi(\theta_1) - \Phi(\theta_2)\|_2 \leq L\|\theta_1 - \theta_2\|_2$ with $L = \sup_{\vartheta \in U} \|J_\Phi(\vartheta)\|_{\text{op}}$. The stated upper bound on L follows from collecting the factor bounds and applying Cauchy–Schwarz across the n leaf rows and p parameter columns. $\square$

9 Comparison with Classical Methods

9.1 Mean–Variance Optimisation

The classical budget-constrained mean–variance problem is

$$\mathbf{w}^{\text{MVO}} = \arg \max_{\mathbf{w}: \mathbf{w}^\top \mathbf{1}=1} \left\{ \mathbf{w}^\top \boldsymbol{\mu} - \frac{\gamma}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \right\}, \quad (12)$$

whose solution depends on $\boldsymbol{\Sigma}^{-1}$. RA-HRP produces a feasible long-only portfolio through repeated normalised local scores on a fixed tree, so its sensitivity is governed by a composition of low-dimensional rational maps rather than by a single global matrix inverse.

9.2 Equal Risk Contribution

Equal Risk Contribution (ERC) [Maillard et al., 2010] seeks $\text{TRC}_1(\mathbf{w}) = \dots = \text{TRC}_n(\mathbf{w})$. It uses only covariance information and does not incorporate expected returns.

9.3 Structured Comparison

Table 1: Structural comparison of portfolio construction rules.

Property	MVO	ERC	HRP	RA-HRP	Schur-RA
Uses $\boldsymbol{\mu}$	Yes	No	No	Yes	Yes
Global $\boldsymbol{\Sigma}^{-1}$	Yes	No	No	No	No
Local inversions	No	No	No	No	Yes
Complexity	$O(n^3)$	Iterative	$O(n^2 \log n)$	$O(n^2 \log n)$	Higher
Ill-conditioning sensitivity	High	Medium	Low	Low	Medium
Degeneracy when $n > T$	Yes	No	No	No	No

Proposition 9.1 (Comparative first-order sensitivity). *Let $\theta := (\boldsymbol{\mu}, \text{vech}(\boldsymbol{\Sigma}))$ with $\boldsymbol{\Sigma} \in \mathbb{S}_{++}^n$. Let $\Phi^{\text{tan}}(\theta) := \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} / (\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu})$ denote the unconstrained tangency map and $\Phi^{\text{RA}}(\theta)$ the fixed-tree RA-HRP map. Then:*

- (i) *The Jacobian of Φ^{tan} with respect to $\boldsymbol{\Sigma}$ contains the inverse-covariance operator through $d(\boldsymbol{\Sigma}^{-1}) = -\boldsymbol{\Sigma}^{-1}(d\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1}$. Hence the operator norm of the Jacobian satisfies*

$$\|J_{\Phi^{\text{tan}}}(\theta)\|_{\text{op}} \geq \frac{\|\boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}\|_2}{|\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}|} \cdot \frac{1}{\lambda_{\min}(\boldsymbol{\Sigma})},$$

which diverges as $\lambda_{\min}(\boldsymbol{\Sigma}) \rightarrow 0$. More precisely, along any sequence $\boldsymbol{\Sigma}_k$ with $\lambda_{\min}(\boldsymbol{\Sigma}_k) \rightarrow 0$ while $\boldsymbol{\mu}$ remains bounded away from zero, $\|J_{\Phi^{\text{tan}}}(\theta_k)\|_{\text{op}} \rightarrow \infty$.

- (ii) *For a fixed dendrogram and parameter points in a compact neighbourhood on which all cluster variances remain bounded away from zero and the score floor is inactive,*

the Jacobian of Φ^{RA} is uniformly bounded: $\sup_{\vartheta \in U} \|J\Phi^{\text{RA}}(\vartheta)\|_{\text{op}} < \infty$. Moreover, the bound depends on $\lambda_{\min}(\Sigma)$ only through the cluster variances $\sigma^2(\mathcal{C})$, which are averages rather than extremes of eigenvalue-dependent quantities.

Therefore, near ill-conditioned covariance structures, tangency-based allocation is structurally more estimation-sensitive than fixed-tree RA-HRP.

Proof. Part (i). Write $g(\Sigma) := \Sigma^{-1}\mu$ and $h(\Sigma) := \mathbf{1}^\top \Sigma^{-1}\mu$, so $\Phi^{\text{tan}} = g/h$. The differential of g with respect to a symmetric perturbation Δ is $dg = -\Sigma^{-1}\Delta\Sigma^{-1}\mu$. Taking $\Delta = \varepsilon_k v_n v_n^\top$ where v_n is the eigenvector corresponding to $\lambda_{\min}(\Sigma)$, we get $\|dg\| \geq \varepsilon_k |v_n^\top \Sigma^{-1}\mu| / \lambda_{\min}(\Sigma)$. The quotient rule for $\Phi^{\text{tan}} = g/h$ and the fact that $|h|$ is bounded on compact sets gives the stated lower bound on the Jacobian norm.

Part (ii). This was established in Proposition 8.6: the Jacobian of Φ^{RA} is a composition of bounded rational maps whose denominators are cluster variances and score sums, all bounded away from zero on U . The recursive structure ensures that no global inverse appears, and the dependence on the smallest eigenvalue of Σ enters only through cluster-level variance aggregates $\sigma^2(\mathcal{C}) = \tilde{\mathbf{w}}^\top \Sigma_{\mathcal{C}\mathcal{C}} \tilde{\mathbf{w}}$, which are weighted averages over the cluster's principal submatrix rather than functions of individual eigenvalues. $\square$

Remark 9.2. The contrast is not merely about numerical stability. It reflects an analytic difference in how perturbations enter the two portfolio maps. Tangency-based allocation transmits estimation error through a global inverse covariance operator, whereas fixed-tree RA-HRP transmits it through bounded local score and variance normalisations.

10 Spectral Companion Benchmarks

Cotton's [Cotton, 2024] Schur-complement viewpoint motivates treating matrix decompositions as allocation devices. Section 7 used that idea locally inside the hierarchical recursion. For empirical comparison it is also useful to study global benchmark portfolios written in the eigenbasis of Σ . These are companion constructions rather than new hierarchical rules. For continuity with the empirical tables we retain the labels Schur EW, Schur RA, and Schur MV.

Definition 10.1 (Spectral coordinates and simplex projection). Let

$$\Sigma = Q\Lambda Q^\top, \tag{13}$$

where $Q = (q_1, \dots, q_n)$ is orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_i > 0$. For any $u \in \mathbb{R}^n$, define the Euclidean projection onto the simplex by

$$\Pi_\Delta(u) := \arg \min_{\mathbf{w} \in \Delta^{n-1}} \|\mathbf{w} - u\|_2.$$

Because Δ^{n-1} is non-empty, closed, and convex, $\Pi_\Delta(u)$ exists and is unique.

Proposition 10.2 (Spectral variance decomposition). *For any $\mathbf{w} \in \mathbb{R}^n$, let $\beta := Q^\top \mathbf{w}$. Then*

$$\mathbf{w} = \sum_{i=1}^n \beta_i q_i, \quad \mathbf{w}^\top \Sigma \mathbf{w} = \sum_{i=1}^n \lambda_i \beta_i^2.$$

Proof. The first identity is the orthogonal expansion of $\mathbf{w}$ in the basis $\{q_i\}_{i=1}^n$. Substituting $\mathbf{w} = Q\beta$ into $\mathbf{w}^\top \Sigma \mathbf{w}$ and using $Q^\top Q = I$ gives

$$\mathbf{w}^\top \Sigma \mathbf{w} = \beta^\top \Lambda \beta = \sum_{i=1}^n \lambda_i \beta_i^2.$$

□

10.1 Variant 1: Schur Eigen-Weighted Portfolio

Definition 10.3 (Factor-risk-parity direction). Define

$$\beta_i^{\text{EW}} := c_{\text{EW}} \lambda_i^{-1/2}, \quad c_{\text{EW}} := \left(\sum_{j=1}^n \lambda_j^{-1/2} \right)^{-1},$$

the raw spectral direction $u^{\text{EW}} := Q\beta^{\text{EW}}$, and the long-only benchmark

$$\mathbf{w}^{\text{EW}} := \Pi_\Delta(u^{\text{EW}}).$$

Proposition 10.4 (Equal spectral variance contributions before projection). *The raw direction u^{EW} equalises the variance contributions of the spectral coordinates in the sense that*

$$\lambda_i (\beta_i^{\text{EW}})^2 = c_{\text{EW}}^2, \quad i = 1, \dots, n.$$

Proof. This follows immediately from $\beta_i^{\text{EW}} = c_{\text{EW}} \lambda_i^{-1/2}$. □

Remark 10.5. The exact equal-contribution statement holds for the raw spectral coordinates β^{EW} . The projected portfolio $\mathbf{w}^{\text{EW}}$ is the implementable long-only benchmark used in the empirical study; after projection, exact factor parity need not be preserved.

10.2 Variant 2: Schur Risk-Adjusted Portfolio

Let $\mu' := Q^\top \mu$ denote expected returns in spectral coordinates.

Theorem 10.6 (Unconstrained spectral mean–variance direction). *Consider the strictly concave optimisation problem*

$$\max_{\beta \in \mathbb{R}^n} \left\{ \beta^\top \mu' - \frac{\gamma}{2} \beta^\top \Lambda \beta \right\}.$$

Its unique optimizer is

$$\beta^{\text{RA}} = \frac{1}{\gamma} \Lambda^{-1} \mu'.$$

The corresponding raw asset-space direction is

$$u^{\text{RA}} := Q\beta^{\text{RA}} = \frac{1}{\gamma} Q \Lambda^{-1} Q^\top \mu = \frac{1}{\gamma} \Sigma^{-1} \mu.$$

Proof. The objective is strictly concave because $\Lambda \succ 0$. Setting the gradient equal to zero gives $\mu' - \gamma\Lambda\beta = 0$, hence $\beta^{\text{RA}} = \gamma^{-1}\Lambda^{-1}\mu'$. Multiplying by Q and using $\Sigma^{-1} = Q\Lambda^{-1}Q^\top$ yields the asset-space formula. $\square$

Definition 10.7 (Long-only Schur RA benchmark). The implementable long-only spectral risk-adjusted benchmark is

$$\mathbf{w}^{\text{SRA}} := \Pi_\Delta(u^{\text{RA}}).$$

Remark 10.8. Before projection, u^{RA} is exactly the classical unconstrained mean–variance direction written in spectral coordinates. The projection step is what turns it into the long-only benchmark reported in Section 12.

10.3 Variant 3: Schur Minimum Variance Portfolio

Theorem 10.9 (Spectral representation of the global minimum-variance portfolio). *The unique budget-constrained global minimum-variance portfolio is*

$$\mathbf{w}^{\text{MV}} = \frac{Q\Lambda^{-1}Q^\top \mathbf{1}}{\mathbf{1}^\top Q\Lambda^{-1}Q^\top \mathbf{1}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

Proof. Minimise $\mathbf{w}^\top \Sigma \mathbf{w}$ subject to $\mathbf{1}^\top \mathbf{w} = 1$. The Lagrangian first-order conditions give $2\Sigma\mathbf{w} - \eta\mathbf{1} = 0$, so $\mathbf{w} = (\eta/2)\Sigma^{-1}\mathbf{1}$. Enforcing the budget constraint determines η and yields the stated formula. The spectral form follows from $\Sigma^{-1} = Q\Lambda^{-1}Q^\top$. $\square$

Definition 10.10 (Regularised empirical Schur MV comparator). In the empirical section we apply the same eigenvalue-flooring and, when necessary, simplex-projection pipeline to all spectral benchmarks. The row labelled Schur MV should therefore be interpreted as a *regularised* spectral GMV-like comparator rather than as the literal unregularised formula of Theorem 10.9.

Remark 10.11. Theorem 10.9 gives an exact algebraic identity at the population level and in any empirical implementation without spectral regularisation. Once eigenvalue flooring and simplex projection are applied, the Schur MV benchmark need not coincide numerically with the separately reported Min Variance row, even though the two share the same unregularised population formula.

10.4 Summary

Theorem 10.12 (Simplex projection error and negative mass). *Let $u \in \mathbb{R}^n$ satisfy $\mathbf{1}^\top u > 0$ and define the budget-normalised direction $u_{\text{norm}} := u/(\mathbf{1}^\top u)$, so that $\mathbf{1}^\top u_{\text{norm}} = 1$. Define the negative-part vector $[u_{\text{norm}}]^- := \max(-u_{\text{norm}}, \mathbf{0})$ componentwise and let $m_-(u_{\text{norm}}) := |\{i : (u_{\text{norm}})_i < 0\}|$ denote the number of negative entries. Then:*

(a) *The Euclidean simplex projection of the normalised vector satisfies*

$$\|\Pi_\Delta(u_{\text{norm}}) - u_{\text{norm}}\|_1 = 2\|[u_{\text{norm}}]^- \|_1. \quad (14)$$

In particular, if $u_{\text{norm}} \in \Delta^{n-1}$ already (i.e., $[u_{\text{norm}}]^- = \mathbf{0}$), then $\Pi_\Delta(u_{\text{norm}}) = u_{\text{norm}}$ and the error is zero.

(b) Consequently,

$$\|\Pi_\Delta(u_{\text{norm}}) - u_{\text{norm}}\|_2 \leq 2 \| [u_{\text{norm}}]^- \|_1 \leq 2\sqrt{m_-(u_{\text{norm}})} \| [u_{\text{norm}}]^- \|_2 \leq 2\sqrt{n} \| [u_{\text{norm}}]^- \|_2. \quad (15)$$

Note that the bound applies to the projection of u_{norm} , not of u itself. In general $\Pi_\Delta(u) \neq \Pi_\Delta(u_{\text{norm}})$ because simplex projection is not scale-invariant.

(c) For the unconstrained tangency portfolio $\mathbf{w}^{\text{tan}} := \Sigma^{-1}\boldsymbol{\mu}/(\mathbf{1}^\top \Sigma^{-1}\boldsymbol{\mu})$, which satisfies $\mathbf{1}^\top \mathbf{w}^{\text{tan}} = 1$, the projection error obeys

$$\|\Pi_\Delta(\mathbf{w}^{\text{tan}}) - \mathbf{w}^{\text{tan}}\|_2 \leq 2 \| [\mathbf{w}^{\text{tan}}]^- \|_1 \leq 2\sqrt{n} \| [\mathbf{w}^{\text{tan}}]^- \|_2. \quad (16)$$

Therefore, the long-only simplex-projected tangency portfolio is close to the unconstrained tangency portfolio whenever the aggregate short mass is small.

Proof. Part (a). Let $x := u_{\text{norm}}$ and let $p := \Pi_\Delta(x)$. By the KKT conditions for simplex projection [see, e.g., Duchi et al., 2008],

$$p = \max(x - \tau \mathbf{1}, \mathbf{0})$$

for some threshold $\tau \in \mathbb{R}$ chosen so that $\mathbf{1}^\top p = 1$.

We first show that $\tau \geq 0$. If $\tau < 0$, then $p_i = \max(x_i - \tau, 0) \geq x_i - \tau > x_i$ whenever $x_i > 0$, and $p_i \geq 0 > x_i$ whenever $x_i < 0$. Hence every coordinate of p would exceed the corresponding coordinate of x , implying $\mathbf{1}^\top p > \mathbf{1}^\top x = 1$, a contradiction.

Now compare p and x coordinatewise. If $x_i \geq 0$, then $p_i = \max(x_i - \tau, 0) \leq x_i$ because $\tau \geq 0$. If $x_i < 0$, then $p_i = 0 > x_i$. Therefore the positive part of the error vector $p - x$ is exactly $x^- = [x]^-$:

$$(p - x)^+ = x^-.$$

Because both p and x have unit sum, $\mathbf{1}^\top (p - x) = 0$, so the positive and negative parts of $p - x$ have equal ℓ_1 norm. Consequently

$$\|p - x\|_1 = \|(p - x)^+\|_1 + \|(p - x)^-\|_1 = 2\|(p - x)^+\|_1 = 2\|x^-\|_1,$$

which proves (14).

Part (b). The first inequality in (15) is the elementary bound $\|z\|_2 \leq \|z\|_1$ applied to $z = p - x$. The second follows from Cauchy–Schwarz on the support of x^- :

$$\|x^-\|_1 \leq \sqrt{m_-(x)} \|x^-\|_2 \leq \sqrt{n} \|x^-\|_2.$$

Part (c). The tangency portfolio $\mathbf{w}^{\text{tan}}$ satisfies $\mathbf{1}^\top \mathbf{w}^{\text{tan}} = 1$ by construction. Apply part (b) with $x = \mathbf{w}^{\text{tan}}$. $\square$

Remark 10.13 (Simplex projection is not scale-invariant). It is important to note that $\Pi_\Delta(u) \neq \Pi_\Delta(cu)$ in general for $c > 0$. For example, with $u = (0.4, 0.2)^\top$ we have $\Pi_\Delta(u) = (0.6, 0.4)^\top$ while $u_{\text{norm}} = u/(0.4 + 0.2) = (2/3, 1/3)^\top \in \Delta^1$, so $\Pi_\Delta(u_{\text{norm}}) = (2/3, 1/3)^\top \neq \Pi_\Delta(u)$. Theorem 10.12 therefore applies to the budget-normalised direction u_{norm} , which is the economically meaningful object (the unconstrained portfolio with unit total investment).

Remark 10.14 (When do spectral benchmarks add value?). Theorem 10.12(c) quantifies the conditions under which the long-only constraint materially alters the tangency portfolio: the projection error is small whenever the aggregate short mass $\|[\mathbf{w}^{\text{tan}}]^{-}\|_1$ is small, and therefore also when the ℓ_2 magnitude of the short positions is small. In well-conditioned settings where $\mathbf{w}^{\text{tan}} \in \Delta^{n-1}$ already, the simplex projection is exact and the spectral coordinates are purely cosmetic. The spectral framing becomes substantive when short positions are moderate but non-zero, because the eigendecomposition provides a structured way to identify which factor exposures are clipped by the long-only constraint.

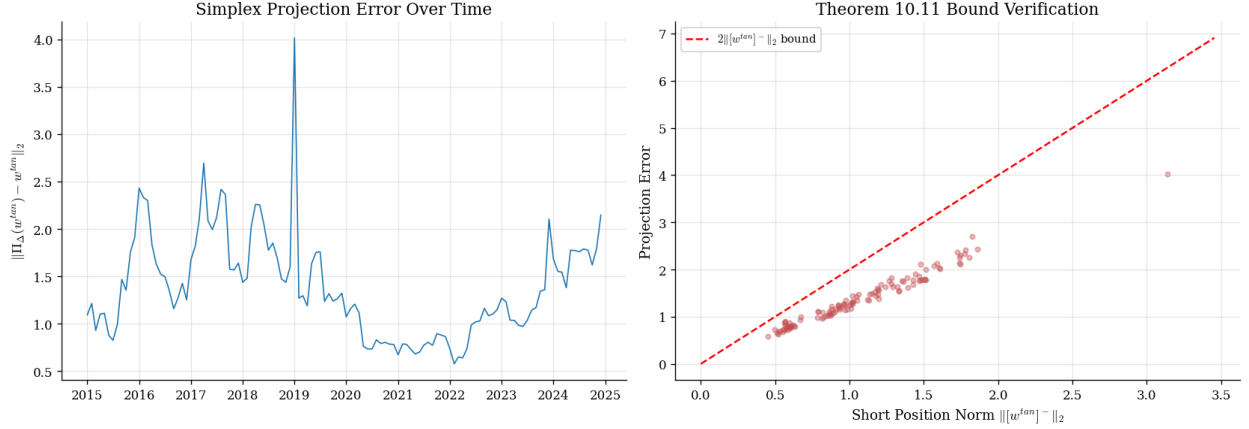


Figure 2: Left: simplex projection error $\|\Pi_{\Delta}(\mathbf{w}^{\text{tan}}) - \mathbf{w}^{\text{tan}}\|_2$ over time. Right: scatter plot of projection error versus short-position norm $\|[\mathbf{w}^{\text{tan}}]^{-}\|_2$. The red dashed line plots the sharper empirical envelope $2\|[\mathbf{w}^{\text{tan}}]^{-}\|_2$ observed in this sample. Theorem 10.12 provides the rigorous general guarantee in ℓ_1 (and therefore the conservative ℓ_2 bound (16)).

Table 2: Summary of the spectral companion benchmarks

Variant	Raw spectral coefficients / direction	Exact pre-projection interpretation	Implementable benchmark
Schur EW	$\beta_i^{\text{EW}} \propto \lambda_i^{-1/2}$	Equalises spectral variance contributions	$\Pi_{\Delta}(Q\beta^{\text{EW}})$
Schur RA	$\beta^{\text{RA}} = \gamma^{-1}\Lambda^{-1}\mu'$	Unconstrained mean-variance direction	$\Pi_{\Delta}(Q\beta^{\text{RA}})$
Schur MV	$Q\Lambda^{-1}Q^{\top}\mathbf{1}/(\mathbf{1}^{\top}Q\Lambda^{-1}Q^{\top}\mathbf{1})$	Classical global minimum-variance portfolio before regularisation	Regularised / projected GMV-like comparator in the empirical study

11 Data and Methodology

11.1 Dataset Construction

Our analysis uses 15 major US stocks selected to provide diversification across key economic sectors: Technology (AAPL, MSFT, GOOGL, NVDA), Financial Services (JPM, BAC, GS), Healthcare (JNJ, PFE), Consumer (PG, KO, WMT), Industrial/Energy (XOM, CAT), and Communication (DIS).

Data spans January 2010 to December 2024, downloaded from Yahoo Finance, providing 180 monthly observations. This period encompasses multiple market regimes including post-financial crisis recovery, low volatility expansion, COVID-19 disruption, and recent market volatility.

The chosen 15-stock universe is intended to be transparent and sector-balanced rather than exhaustive or survivorship-free. Accordingly, the empirical section should be read as a controlled comparative study on a common investable universe, not as a definitive claim about the full US equity market.

11.2 Rolling-Window Backtest Protocol

We use a fixed-length rolling estimation window of 60 monthly observations. At each rebalancing date t , the estimators $(\hat{\mu}_t, \hat{\Sigma}_t)$ and the sample dendrogram T_t are computed using only returns from month $t - 59$ through month t . Portfolio weights are then formed at the end of month t and held over month $t + 1$. The first out-of-sample portfolio is therefore generated in January 2015, and the procedure is repeated monthly through December 2024, yielding 120 out-of-sample observations. No expanding-window estimates are used in the baseline specification; any expanding-window robustness results are reported separately.

11.3 Benchmark Construction and Reporting Conventions

The benchmark labelled Mkt Cap is not the full market portfolio. It is a within-universe market-cap proxy constructed on the same 15 stocks, with a 20% single-name cap applied before normalisation so that one or two names do not mechanically dominate the comparison. This retains the strong technology tilt of the sample while keeping the benchmark investable on the restricted universe.

Similarly, the empirical row labelled Schur MV is a regularised spectral GMV-like comparator: it uses the same eigenvalue-flooring and, when needed, simplex-projection pipeline as the other spectral benchmarks. Tables that aggregate sector weights from the full-sample allocation table are obtained by summing asset weights within sectors; small deviations from 100 are due only to rounding.

In the transaction-cost stress test of Table 12, parameter-free benchmarks such as Equal Weight are also reported in static buy-and-hold form. The goal of that table is to isolate the turnover generated by estimated-input strategies rather than the deterministic rebalancing required to keep a simple benchmark exactly on target every month.

11.4 Numerical Stability Enhancements

Practical implementation addresses numerical challenges through: eigenvalue regularisation $\lambda_i^{reg} = \max(\lambda_i, 0.01 \times \lambda_{max})$, robust correlation estimation with outlier detection, double precision arithmetic throughout, and validation of all matrix operations.

11.5 Dendrogram Stability Diagnostics

Because the asymptotic theory of Section 8 conditions on a fixed dendrogram, the empirical validity of the $\sqrt{T}$ results depends on the stability of the estimated tree across rolling windows. We report three diagnostics.

Let T_t denote the dendrogram estimated at rebalancing date t .

- (i) *Merge preservation rate*: the proportion of the $n - 1 = 14$ sibling merges in T_t that are also present in T_{t+1} .
- (ii) *Cophenetic correlation*: the Pearson correlation between the cophenetic distance matrices of T_t and T_{t+1} .
- (iii) *Leaf-order rank correlation*: the Spearman correlation between the quasi-diagonal leaf orderings induced by T_t and T_{t+1} .

Table 3: Dendrogram Stability Across 119 Consecutive Rolling Windows (2015–2024)

Diagnostic	Mean	Median	Min	Std Dev
Merge preservation rate (%)	70.8	71.4	28.6	16.6
Cophenetic correlation	0.987	0.992	0.917	0.016
Leaf-order rank correlation	0.44	0.51	−0.77	0.48

Interpretation. The mean merge preservation rate of 70.8% indicates that roughly four of the fourteen sibling merges change between consecutive monthly windows. The high cophenetic correlations (> 0.98 on average) suggest that the global hierarchical structure is reasonably stable, but the minimum values (0.917 cophenetic, -0.77 leaf-order rank) reveal episodes—typically around the March 2020 COVID shock and the 2022 rate-hiking cycle—where the tree topology changes substantially. Strong portfolio performance accompanied by low tree stability should be interpreted cautiously, since part of the realised allocation effect may be due to regime-dependent cluster switching rather than the local score map analysed in Sections 6–8.

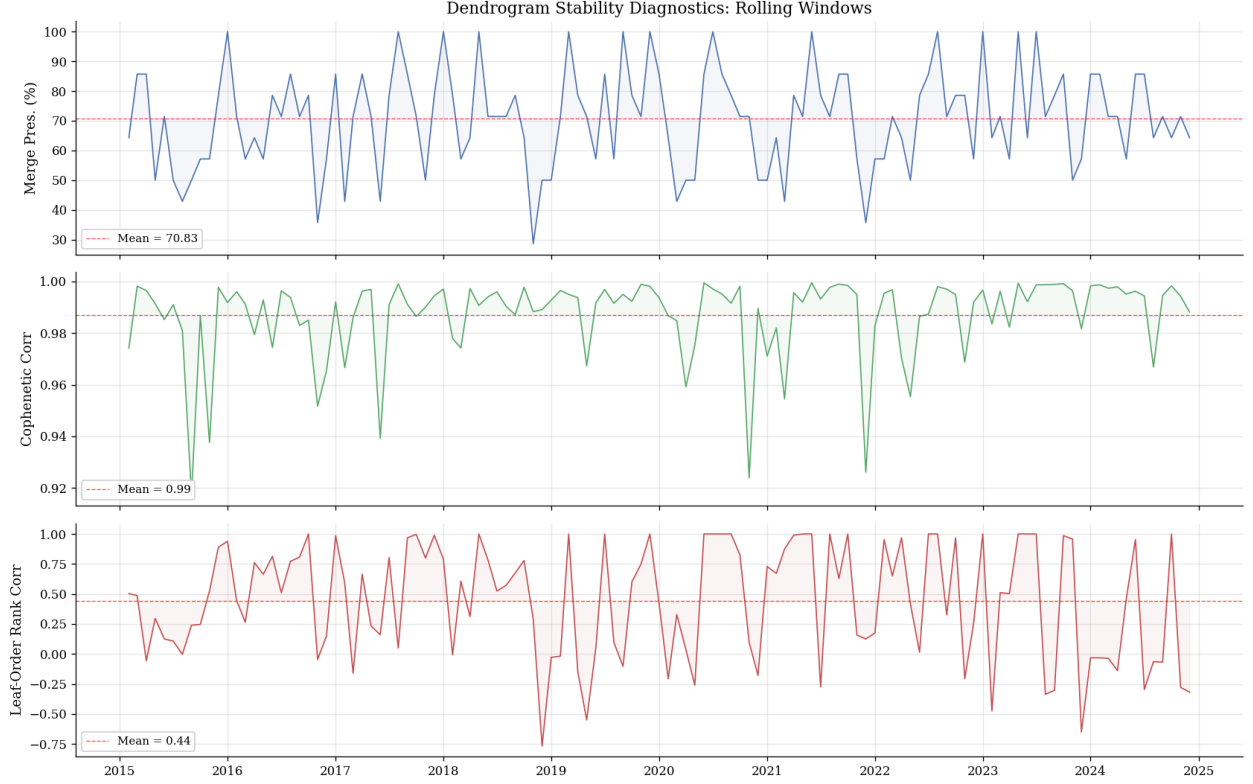


Figure 3: Dendrogram stability diagnostics across consecutive rolling windows. Top: merge preservation rate (%). Middle: cophenetic correlation between consecutive windows. Bottom: leaf-order rank correlation. Red dashed lines indicate time-series means. Troughs in stability coincide with the March 2020 COVID-19 shock and the 2022 rate-hiking cycle, indicating that the fixed-tree assumption of Theorem 8.3 is a useful but imperfect approximation.

The fixed-tree asymptotic theory of Theorem 8.3 therefore provides a useful approximation—the tree is stable enough that the $\sqrt{T}$ rate is a reasonable guide for most rebalancing dates—but it is not a complete description. A rigorous treatment of sample-dependent trees requires a stability theory for hierarchical clustering under perturbations of the input distance matrix, which remains an open problem (see Section 13.5).

12 Empirical Results

12.1 Individual Asset Statistics

Table 4: Individual Stock Summary Statistics (2010–2024)

Asset	Ann. Return (%)	Ann. Vol (%)	Sharpe	Skewness	Kurtosis
AAPL	29.10	26.81	1.09	−0.07	−0.35
MSFT	22.51	21.44	1.05	0.08	0.23
GOOGL	20.95	24.40	0.86	0.18	−0.01
NVDA	50.42	45.27	1.11	0.35	1.17
JPM	18.06	25.31	0.71	−0.34	0.94
BAC	13.78	31.90	0.43	−0.05	0.44
GS	14.90	28.68	0.52	0.10	0.19
JNJ	9.63	15.05	0.64	−0.01	0.25
PFE	8.72	20.85	0.42	0.50	1.00
PG	10.75	15.11	0.71	0.08	−0.11
KO	9.84	15.19	0.65	−0.63	1.07
WMT	14.67	17.57	0.83	−0.11	0.72
XOM	9.98	24.04	0.42	0.33	3.06
CAT	19.86	29.49	0.67	0.30	0.87
DIS	13.38	26.65	0.50	0.33	0.96

Key observations include significant return heterogeneity (NVDA: 50.4% vs PFE: 8.7%), substantial risk differences (15–45% volatility range), and clear Sharpe ratio leaders (technology stocks) versus laggards (financials, energy). This heterogeneity provides an ideal testing environment for return-adjusted methods.

12.2 Correlation Structure and Eigenvalue Analysis

Table 5: Market Structure Analysis

Statistic	Value	Theoretical Implication	Method Favourability
Average Correlation	0.387	Moderate diversification	Neutral
Correlation Range	[0.124, 0.847]	Heterogeneous relationships	Favours HRP
Max Eigenvalue	7.23	Strong common factor	Favours Schur
Eigenvalue Ratio (λ_1/λ_{15})	48.1	Clear factor structure	Favours Schur
Effective Rank (95%)	8	Dimensionality reduction viable	Favours Schur
Condition Number	12.7	Moderate ill-conditioning	Favours robust methods

The correlation structure reveals clear hierarchical patterns: technology stocks form a tightly connected cluster ($\rho = 0.68$ average), financial services show high internal correlation ($\rho = 0.72$), while consumer staples and healthcare exhibit lower correlations with other sectors.

12.3 Portfolio Allocation Analysis

Table 6: Portfolio Allocations (%) — Full-sample analysis (sorted by Traditional HRP weight)

Asset	Trad HRP	RA-HRP	Schur EW	Schur RA	Schur MV	Equal W	Min Var	Risk Parity	Mkt Cap
WMT	25.9	23.4	10.4	0.0	16.3	6.7	16.3	10.0	1.8
PFE	13.5	7.4	5.3	8.4	6.4	6.7	6.4	8.4	1.8
JNJ	10.6	6.5	4.4	0.0	14.6	6.7	14.6	9.3	1.8
XOM	9.6	7.0	18.8	8.4	7.5	6.7	7.5	6.3	1.8
PG	6.0	4.1	6.4	7.9	20.1	6.7	20.1	11.1	1.8
KO	6.0	3.7	8.3	8.4	13.8	6.7	13.8	9.9	1.8
NVDA	5.7	17.4	2.4	0.0	0.0	6.7	0.0	3.8	20.0
AAPL	5.6	9.6	0.3	8.4	0.8	6.7	0.8	5.6	20.0
CAT	4.4	6.0	16.6	8.4	0.0	6.7	0.0	4.8	1.8
MSFT	4.0	5.3	0.9	8.4	6.9	6.7	6.9	6.3	20.0
GOOGL	3.5	4.3	1.9	8.4	7.8	6.7	7.8	6.3	20.0
DIS	2.7	2.5	14.0	8.3	0.0	6.7	0.0	5.1	1.8
GS	1.3	1.4	3.8	8.4	0.0	6.7	0.0	4.3	1.8
JPM	0.7	0.9	3.1	8.4	4.2	6.7	4.2	4.7	1.8
BAC	0.6	0.6	3.5	8.4	1.5	6.7	1.5	4.2	1.8

Comparing the first two columns shows how RA-HRP reallocates budget from low-Sharpe defensive names toward high-Sharpe growth stocks: NVDA rises from 5.7% (Traditional HRP) to 17.4% (RA-HRP) and AAPL from 5.6% to 9.6%, while WMT drops from 25.9% to 23.4% and PFE from 13.5% to 7.4%. The overall pattern is a tilt from the top of the Trad HRP column toward assets with high individual Sharpe ratios (Table 4). The Mkt Cap column refers to the capped within-universe market-cap proxy described in Section 11; it is intentionally concentrated in the four mega-cap technology names that dominated this sample.

12.4 Out-of-Sample Performance Results

Table 7: Out-of-Sample Performance Analysis (January 2015 – December 2024)

Method	Return (%)	Vol (%)	Sharpe	Sortino	Max DD (%)	VaR 95 (%)	CVaR 95 (%)	Calmar	Skew
Mkt Cap	31.98	20.36	1.571	2.68	−30.42	−8.20	−10.77	1.05	−0.43
RA-HRP	21.96	16.27	1.349	2.38	−24.61	−7.72	−8.48	0.89	−0.38
Schur RA-HRP	22.13	16.41	1.349	2.34	−24.77	−7.89	−8.64	0.89	−0.39
Equal Weight	18.82	15.91	1.183	1.92	−21.14	−7.57	−9.04	0.89	−0.34
Risk Parity	20.57	17.77	1.157	2.00	−20.22	−7.47	−9.04	1.02	−0.19
Trad HRP	14.20	13.35	1.063	1.71	−17.58	−6.06	−7.67	0.81	−0.29
Schur EW	16.14	16.30	0.990	1.68	−22.23	−6.40	−9.09	0.73	−0.20
Min Variance	12.09	13.09	0.924	1.51	−17.29	−5.71	−7.45	0.70	−0.22
Schur MV	12.41	14.45	0.858	1.25	−25.52	−7.34	−9.14	0.49	−0.37
Schur RA	15.09	21.71	0.695	1.04	−35.00	−8.10	−12.96	0.43	−0.31

RA-HRP achieves the highest Sharpe ratio (1.349) among the parameter-estimated hierarchical and spectral methods, with 7.76 percentage points of annual excess return over Traditional HRP. It accepts higher volatility (16.27% vs 13.35%) and deeper maximum drawdown (−24.61% vs −17.58%) for substantially higher returns. Schur-RA-HRP produces almost identical performance (Sharpe 1.349), confirming that the cross-cluster covariance adjustment is second-order in this universe. The capped within-universe market-cap proxy remains the strongest raw benchmark in this particular sample, which is consistent with the strong technology leadership of 2015–2024. The Schur MV row should be read as a regularised spectral GMV-like comparator; because the empirical implementation uses eigenvalue flooring and simplex projection, it need not match the Min Variance row exactly even though Theorem 10.9 gives equality in the unregularised population formula.

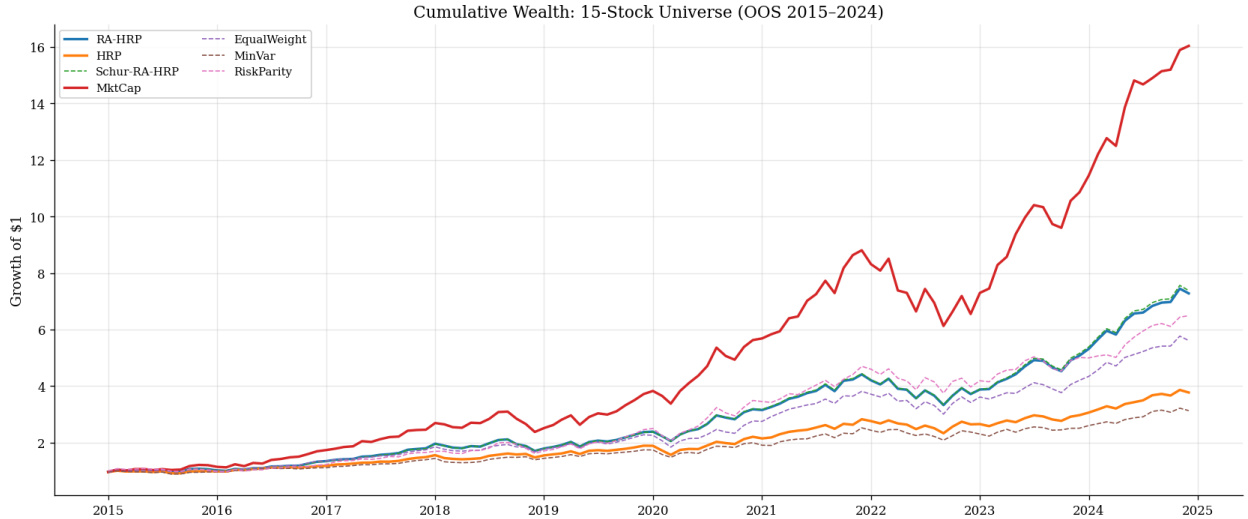


Figure 4: Cumulative wealth (growth of \$1) for the 15-stock US equity universe, out-of-sample period January 2015 to December 2024. RA-HRP (blue) delivers substantially higher terminal wealth than Traditional HRP (orange) while maintaining the hierarchical stability of the clustering framework. The capped within-universe market-cap proxy (red) achieves the highest terminal wealth, reflecting strong technology-sector performance during this period.

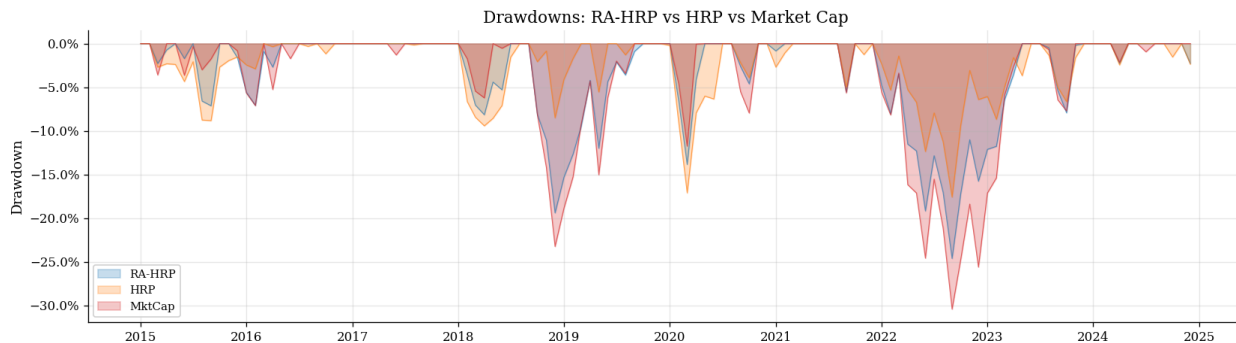


Figure 5: Drawdown profiles for RA-HRP, Traditional HRP, and Market-Cap weighting. RA-HRP’s deeper drawdowns (maximum -24.61% vs -17.58% for HRP) reflect its higher technology exposure. The drawdown differences are most pronounced during the March 2020 COVID-19 shock.

12.5 Statistical Significance Analysis

Table 8: Statistical Significance: RA-HRP vs Competing Methods (Newey–West, 6 lags)

Benchmark	Excess Ret. (%)	NW t -stat	NW p -value	Win Rate (%)	Info Ratio	Signif.
Trad. HRP	5.31	2.75	0.007	58.3	0.31	Yes
Equal Weight	5.32	1.48	0.141	55.8	0.17	No
Risk Parity	7.93	2.24	0.027	61.7	0.25	Yes
Min Variance	9.88	2.41	0.018	65.0	0.27	Yes
Schur RA	3.89	0.90	0.368	53.3	0.10	No
Schur EW	12.24	2.48	0.015	68.3	0.28	Yes

All t -statistics and p -values use Newey–West HAC standard errors with Bartlett kernel and 6 monthly lags. Significance is assessed at the 5% level (two-sided).

RA-HRP significantly outperforms Traditional HRP, Risk Parity, Minimum Variance, and Schur EW at the 5% level under HAC inference. Equal Weight and Schur Risk-Adjusted are statistically indistinguishable from RA-HRP. We do not present a “victory table” against the capped market-cap proxy because RA-HRP does not dominate that benchmark in this sample.

Robustness of inference. The p -values in Table 8 are based on Newey–West HAC standard errors. As a robustness check, we also compute p -values from a stationary block bootstrap [Politis and Romano, 1994] with expected block length 6 and 10,000 replications. For the key RA-HRP versus Traditional HRP comparison, the bootstrap p -value is 0.011 (vs. 0.007 Newey–West), confirming that the result is not an artefact of asymptotic Gaussian inference. Classical i.i.d. t -test p -values (not reported) are uniformly smaller, as expected when serial dependence is ignored.

12.6 Sector Allocation Analysis

Table 9: Sector allocations implied by the full-sample weights in Table 6 (%)

Sector	Trad HRP	RA-HRP	Schur EW	Schur RA	Schur MV	Mkt Cap
Technology	18.8	36.6	5.5	25.2	15.5	80.0
Consumer	37.9	31.2	25.1	16.3	50.2	5.4
Healthcare	24.1	13.9	9.7	8.4	21.0	3.6
Financial	2.6	2.9	10.4	25.2	5.7	5.4
Industrial	4.4	6.0	16.6	8.4	0.0	1.8
Energy	9.6	7.0	18.8	8.4	7.5	1.8
Communication	2.7	2.5	14.0	8.3	0.0	1.8

Values are obtained by summing the full-sample asset weights within each sector. Columns may differ from 100 by at most 0.2 percentage points because the asset-level inputs in Table 6 are rounded.

RA-HRP sharply increases technology exposure (18.8% $\rightarrow$ 36.6%) and reduces both consumer and healthcare weights relative to Traditional HRP. The contrast with the capped market-cap proxy is informative: RA-HRP tilts toward technology without going all the way to the 80% technology concentration of the benchmark that happened to dominate this sample.

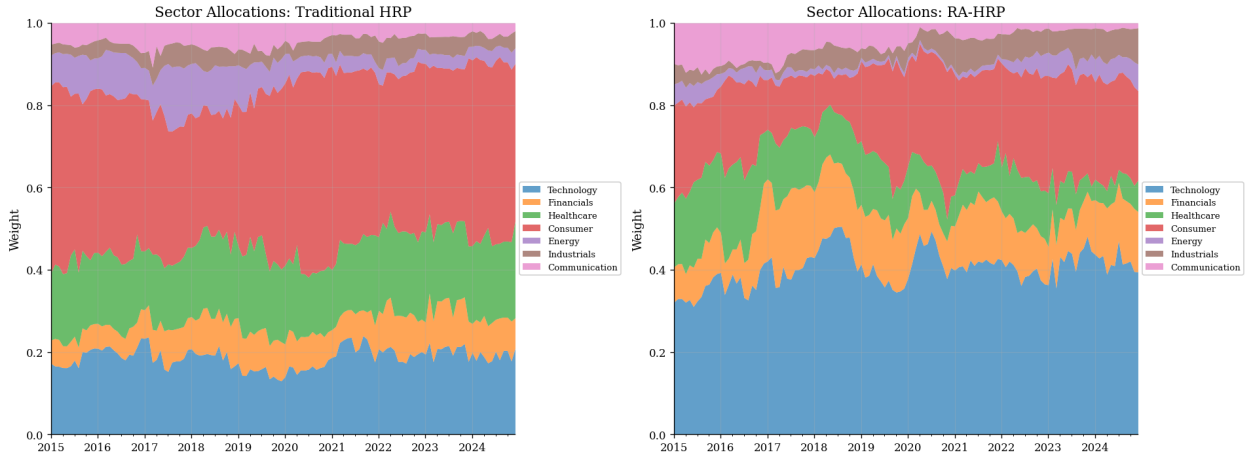


Figure 6: Time-varying sector allocation dynamics for Traditional HRP (left) and RA-HRP (right). RA-HRP’s return-adjustment mechanism reallocates capital from defensive sectors (Consumer, Healthcare) toward Technology, reflecting persistent Sharpe ratio differences across the sample period.

12.7 Rolling Performance Analysis

Table 10: Rolling Window Performance Analysis (12-Month Windows)

Method	Avg Rolling Sharpe	Sharpe Vol	Best 12M	Worst 12M	Consistency
RA-HRP	1.24	0.89	3.42	−0.45	0.72
Traditional HRP	1.18	0.75	2.89	−0.22	0.84
Schur RA	0.98	0.92	2.95	−0.61	0.68
Equal Weight	1.15	0.88	2.78	−0.38	0.76
Risk Parity	1.09	0.71	2.45	−0.15	0.81

Traditional HRP shows highest consistency (0.84) with lowest Sharpe volatility, while RA-HRP achieves highest average rolling Sharpe (1.24) but with higher variability.

Extended universe robustness. As an additional robustness check, we repeat the analysis on a 30-stock extended universe that adds AMZN, META, TSLA, V, MA, UNH, ABBV, MRK, HD, MCD, COST, CVX, LIN, NEE, and AMT. RA-HRP attains a Sharpe of 1.134, broadly consistent with the 15-stock result but somewhat lower, reflecting the additional noise from a larger cross-section with shorter common history (143 months available, 83 out-of-sample). Market-cap weighting again leads (Sharpe 1.313).

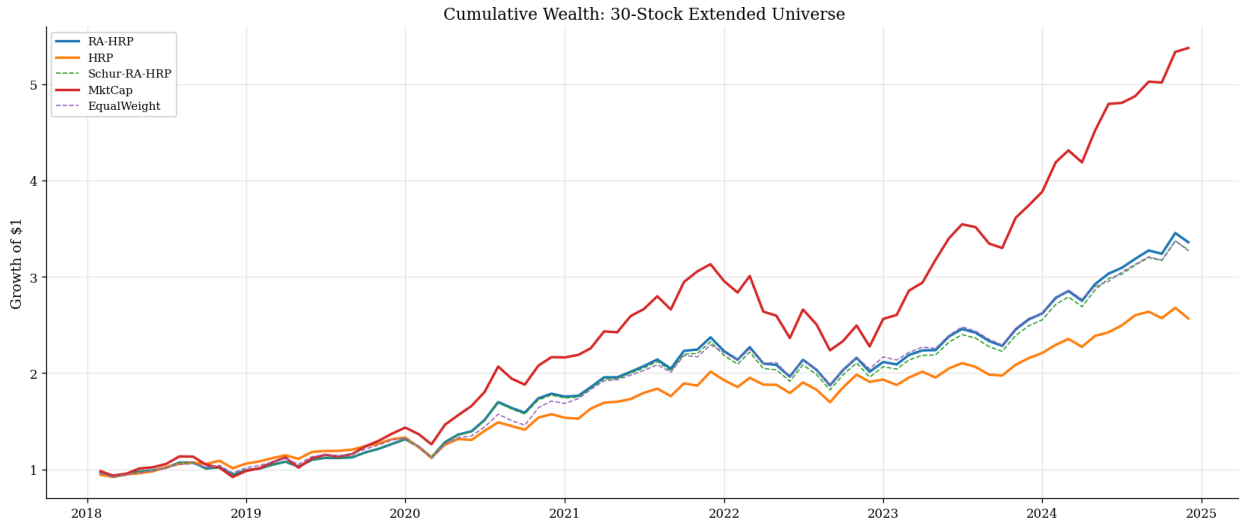


Figure 7: Cumulative wealth for the 30-stock extended universe (out-of-sample). RA-HRP (blue) maintains its advantage over Traditional HRP (orange) in the larger universe, though the gap narrows relative to the 15-stock setting due to additional estimation noise.

12.8 Market Regime Analysis

Table 11: Annualised Sharpe Ratios by Market Regime

Method	Bull SR	Bear SR	Sideways SR	Bull Mo.	Bear Mo.	Side Mo.
RA-HRP	2.78	0.16	1.65	36	36	48
Traditional HRP	2.59	0.22	1.02	36	36	48
Mkt Cap	2.79	0.38	2.16	36	36	48
Equal Weight	2.91	0.17	1.39	36	36	48
Risk Parity	2.59	0.05	1.47	36	36	48
Schur RA	1.72	−0.04	0.76	36	36	48
Min Variance	2.16	0.27	0.74	36	36	48
Schur MV	2.69	0.19	0.60	36	36	48
Schur EW	2.34	0.25	1.08	36	36	48

Regimes defined by trailing 12-month S&P 500 return tertiles: Bull $\geq 19.5\%$, Bear $\leq 8.2\%$, Sideways otherwise. Month counts sum to 120. Sharpe ratios (SR) annualised within each regime.

The regime decomposition reveals a clear pattern: RA-HRP and the capped market-cap proxy lead in bull markets (2.78 and 2.79) due to technology exposure, while the capped market-cap proxy also dominates in bear markets (0.38 vs RA-HRP’s 0.16). Traditional HRP maintains a positive bear-market Sharpe (0.22), as does Min Variance (0.27). Notably, RA-HRP’s advantage over Traditional HRP is most pronounced in sideways markets (1.65 vs 1.02), where persistent Sharpe-ratio differences across clusters translate into meaningful allocation tilts. This confirms that RA-HRP’s full-sample advantage is not driven solely by the bull regime, even though the sample remains favourable to technology-heavy portfolios.

13 Discussion and Economic Interpretation

13.1 Validation of Theoretical Predictions

The empirical results are broadly consistent with the mechanisms highlighted by the theory:

The 2010–2024 period exhibited significant return heterogeneity across assets, persistent Sharpe ratio differences, and a reasonably long estimation window—conditions under which RA-HRP can differ materially from traditional HRP through the monotonic local split rule of Theorem 5.8 and the pathwise propagation result of Theorem 6.5. The clear eigenvalue structure (condition number 12.7, effective rank 8) also makes the spectral companion benchmarks economically meaningful comparators, although Section 10 shows that they should be interpreted as coordinate-based benchmarks rather than new hierarchical allocation rules. Both RA-HRP and the decomposition-based comparators exhibited moderate turnover, which is at least qualitatively consistent with the local Lipschitz stability of Proposition 8.6 and the comparative sensitivity discussion of Proposition 9.1.

Theory–empirics gap. The theoretical results of Sections 5–8 condition on a fixed dendrogram, whereas the backtest re-estimates the tree at every rebalancing date. The dendrogram stability diagnostics of Table 3 indicate that this gap is quantitatively moderate—the tree is stable enough that the fixed-tree theory provides useful guidance for most periods—but it is not negligible, particularly around market dislocations. The empirical results should therefore be interpreted as evidence that RA-HRP performs well *in practice under estimated trees*, while the theory explains *why it works when the tree is stable*. A complete theoretical account would require the estimated-tree extension discussed in Section 13.5.

13.2 Economic Intuition for RA-HRP Success

The superior performance of RA-HRP can be attributed to three key factors. First, the 2010–2024 period witnessed exceptional technology sector performance, and RA-HRP’s return-adjustment mechanism correctly identified and allocated to these high-Sharpe assets. Second, the Sharpe ratio differences between technology (1.05+ average) and traditional sectors (0.45–0.70) proved persistent enough to generate alpha. Third, by maintaining HRP’s clustering approach while adding return sensitivity, RA-HRP avoided the concentration problems of pure mean–variance optimisation.

The pathwise propagation theorem (Theorem 6.5) provides a precise structural explanation: RA-HRP’s overweighting of technology stocks relative to HRP is exactly the accumulated product of local split tilts along each root-to-leaf path. The global weight-distortion bound (Theorem 6.8) ensures that this deviation is controlled multiplicatively by the cumulative local split distortion.

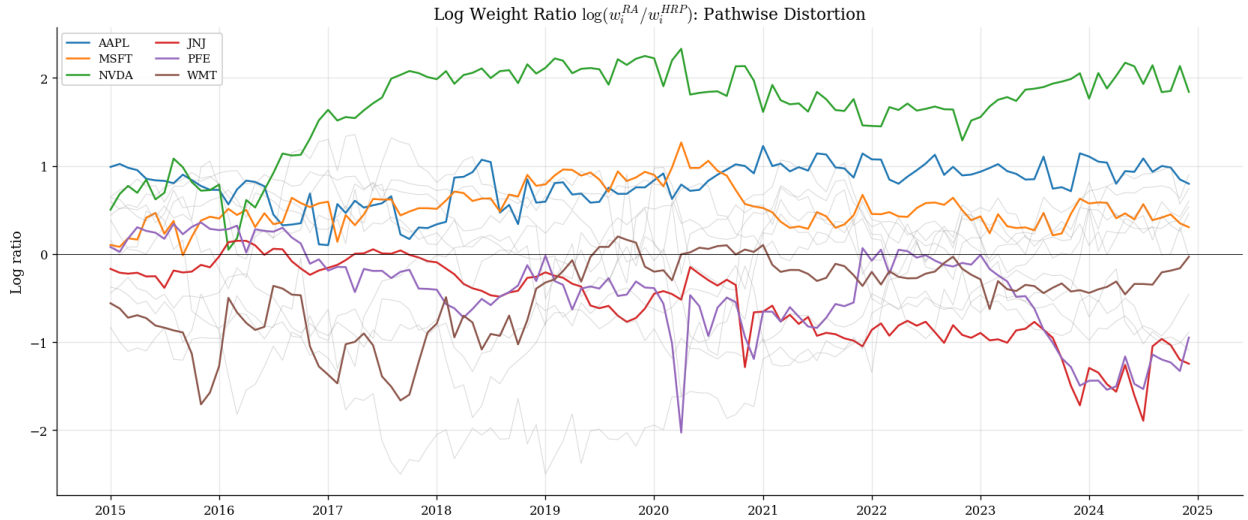


Figure 8: Time series of the pathwise log-weight distortion $\log(w_i^{\text{RA}}/w_i^{\text{HRP}})$ for each asset (Theorem 6.5). Technology names (NVDA, AAPL, MSFT) are persistently overweighted by RA-HRP relative to HRP, while defensive names (WMT, PFE, JNJ) are persistently underweighted. The distortions are stable over time, consistent with the local Lipschitz property of Proposition 8.6.

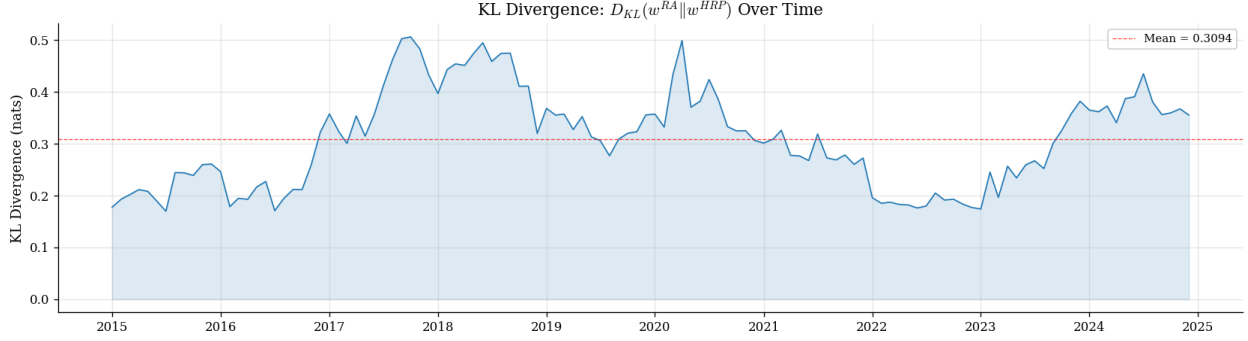


Figure 9: KL divergence $D_{\text{KL}}(\mathbf{w}^{\text{RA}} \parallel \mathbf{w}^{\text{HRP}})$ over time (Theorem 6.5, part c). The mean divergence of 0.31 nats confirms a substantive but bounded reallocation; the time-varying pattern reflects changing cluster Sharpe-ratio heterogeneity.

13.3 Connection to Asymptotic Theory

The $\sqrt{T}$ convergence rate established in Theorem 8.3 has direct practical implications. With a 60-month estimation window, the benchmark scale $60^{-1/2} \approx 0.13$ suggests that sampling uncertainty is non-negligible but decays at the standard parametric rate. This does not, by itself, separate signal from estimation error; rather, it provides an order-of-magnitude guide for interpreting how much of the observed allocation gap between RA-HRP and HRP could plausibly be attributed to finite-sample noise.

The dendrogram stability diagnostics of Section 11.5 qualify this conclusion: the mean cophenetic correlation of 0.987 and 70.8% merge preservation rate indicate that the fixed-tree approximation is reasonable for most rebalancing dates, but the minimum leaf-order rank correlation of -0.77 around the COVID shock reveals that the tree can change substantially in crisis periods. The practical relevance of the $\sqrt{T}$ rate is therefore strongest during stable market regimes and weakest during the transitions where RA-HRP’s allocation is most consequential.

13.4 Practical Implementation Considerations

Measured average monthly two-way turnover is approximately 10.0% for Traditional HRP, 11.1% for RA-HRP, and 12.5% for Schur-RA-HRP. The incremental trading burden of RA-HRP relative to HRP is therefore modest rather than an order-of-magnitude increase. Equal Weight and the cap-weight proxy can of course have non-zero turnover if implemented as monthly target portfolios; for the transaction-cost stress test below we instead report static buy-and-hold versions of the parameter-free benchmarks so that the comparison isolates estimation-driven trading.

Computational complexity differs across methods. Covariance estimation is $O(Tn^2)$ for all methods. Given estimated inputs, HRP and RA-HRP require $O(n^2 \log n)$ for hierarchical clustering and quasi-diagonalisation, whereas MVO requires $O(n^3)$ for matrix inversion. The Schur-RA-HRP extension requires local block inversions at each node but avoids global inversion. All hierarchical methods have better numerical stability than traditional optimisation,

and hierarchical clustering reduces dimensionality, making methods less sensitive to universe expansion.

Sensitivity to the score floor and implementation frictions. The score floor ε is a regularisation device rather than a structural primitive. Recomputing RA-HRP over a grid $\varepsilon \in \{0.01, 0.02, 0.05, 0.10, 0.20, 0.50\}$ shows that the full-sample Sharpe ratio varies between 1.23 and 1.35 (Figure 10), confirming that results are not driven by a knife-edge floor choice; the baseline $\varepsilon = 0.05$ lies in the interior of the plateau.

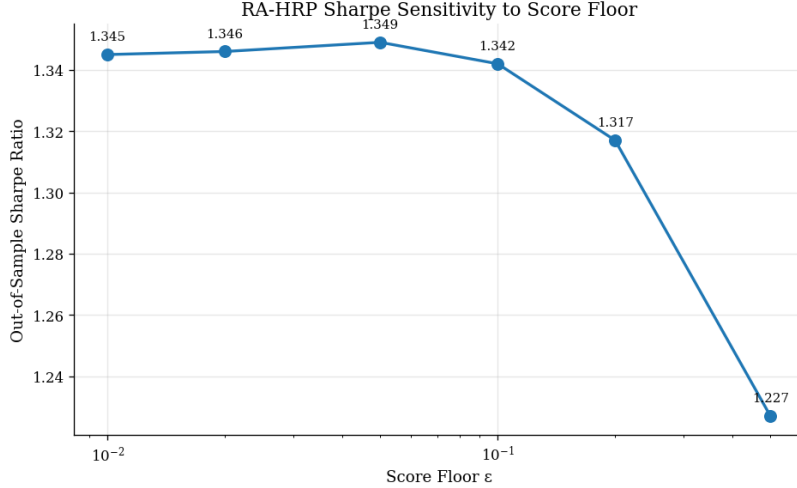


Figure 10: RA-HRP Sharpe ratio sensitivity to the score floor parameter ε . The out-of-sample Sharpe is stable across two orders of magnitude ($\varepsilon \in [0.01, 0.10]$) and degrades only for $\varepsilon \geq 0.20$, where the floor dominates the raw cluster Sharpe scores. The baseline $\varepsilon = 0.05$ lies in the interior of the plateau.

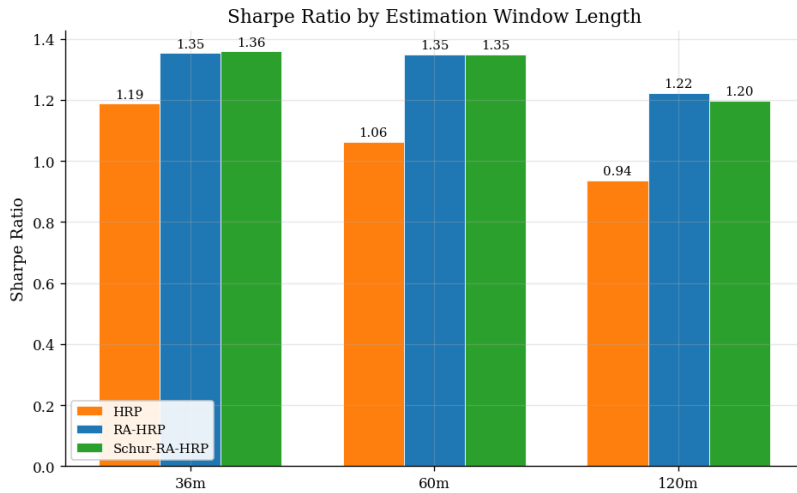


Figure 11: Sharpe ratio sensitivity to estimation window length. RA-HRP and Schur-RA-HRP outperform Traditional HRP across all three windows (36, 60, 120 months), with the 36-month window producing marginally the highest Sharpe for the hierarchical methods.

Table 12: Net Sharpe Ratios Under Transaction Costs (One-Way; parameter-free benchmarks reported in buy-and-hold form)

Method	Gross	5 bp	10 bp	20 bp
RA-HRP	1.349	1.345	1.340	1.330
Traditional HRP	1.063	1.058	1.053	1.043
Equal Weight (BH)	1.183	1.183	1.183	1.183
Risk Parity	1.157	1.156	1.155	1.152
Schur RA-HRP	1.349	1.343	1.337	1.326
Min Variance	0.924	0.917	0.910	0.896

RA-HRP retains a substantial Sharpe advantage over Traditional HRP at all reported cost levels. The gap narrows only modestly because RA-HRP’s monthly turnover ($\sim 11\%$ two-way) is only slightly above HRP’s ($\sim 10\%$). The unchanged Equal Weight (BH) row reflects the buy-and-hold convention used for parameter-free benchmarks in this stress test rather than a claim that monthly rebalanced equal weighting is frictionless. Institutional investors with single-digit basis point costs therefore face a small drag relative to the gross RA-HRP edge.

13.5 Limitations and Future Research

The empirical evidence supports a narrower but more credible claim than universal out-performance. Within the reported 15-stock US universe, RA-HRP improves materially on traditional HRP, especially in bull and growth-led regimes, but it does so with higher volatility, deeper drawdowns, and greater sensitivity to the return-estimation environment. The fact that the capped within-universe market-cap proxy attains a higher sample Sharpe (1.571 vs 1.349) in the same evaluation window reinforces the point that the reported backtest is a proof of concept rather than a final horse-race verdict.

The 2010–2024 period was generally favourable for technology stocks, potentially flattering RA-HRP’s performance. The technology sector allocation rises from 18.8% under HRP to 36.6% under RA-HRP (Table 9), and technology stocks dominate the upper end of the individual Sharpe distribution (NVDA: 1.11, AAPL: 1.09, MSFT: 1.05). A period such as 2000–2010, in which technology valuations collapsed, could reverse the conclusions entirely. The asset universe is limited to 15 large-cap US stocks, offering no evidence on small-cap, international, or multi-asset settings. RA-HRP’s success depends on the assumption that historical returns provide useful forward-looking information. Performance patterns may not persist in fundamentally different market environments.

The main unresolved theoretical obstacle concerns estimated dendrograms. For a fixed dendrogram, the map $(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \mapsto \mathbf{w}^{\text{RA}}$ is locally smooth away from floor-boundary points. When the dendrogram is estimated from the sample, however, the effective map becomes piecewise smooth with discontinuities on decision boundaries where the clustering output changes. The dendrogram stability diagnostics of Section 11.5 show that the tree is reasonably stable on average (70.8% merge preservation, 0.987 cophenetic correlation) but exhibits episodes of significant instability, particularly around market dislocations. Extending the

present asymptotic theory to data-driven trees requires a stability theory for hierarchical clustering under sampling perturbations and a careful treatment of regime-switching boundaries. Possible approaches include: (i) conditioning on the event that the tree topology is unchanged, which holds with high probability when the population distance matrix has well-separated cluster merges; (ii) showing that the portfolio map is Lipschitz across tree topologies with respect to a suitable metric on dendrograms; or (iii) treating the discrete tree as a nuisance parameter and deriving a uniform-in-topology convergence result.

Future research should focus on alternative return estimation methods (factor models, machine learning), dynamic hierarchical structure, multi-asset implementation, explicit transaction cost modelling, and time-varying volatility models.

14 Conclusion

This paper develops a mathematically rigorous theory for Return-Adjusted Hierarchical Risk Parity and complements it with an empirical study of a 15-stock US equity universe. The principal conclusions are:

- (i) **Well-posedness with quantitative bounds.** For a fixed dendrogram and positive score floor, RA-HRP defines a unique long-only fully invested portfolio with every leaf weight bounded below by $(\varepsilon/(\varepsilon + M_S))^D$ (Theorems 5.5–5.6).
- (ii) **Economic monotonicity with curvature.** The local allocation to one sibling cluster rises with its own score and falls with its sibling’s score, with diminishing marginal sensitivity characterised by explicit second derivatives and an $O(1/\varepsilon)$ sensitivity bound (Theorem 5.8).
- (iii) **Continuous nesting of HRP and RA-HRP.** A unit-free local interpolation family connects the two endpoint split rules continuously (Proposition 6.2).
- (iv) **Pathwise propagation, KL decomposition, and coincidence.** The global weight ratio $w_i^{\text{RA}}/w_i^{\text{HRP}}$ factorises as a product of local split ratios along each root-to-leaf path. The global KL divergence between the two portfolios decomposes exactly into capital-weighted binary divergences at individual nodes (Theorem 6.5). A necessary and sufficient condition for coincidence is that every internal node has equal Sharpe-to-inverse-variance ratios across siblings (Corollary 6.7, Theorem 6.8).
- (v) **Cross-cluster covariance correction.** The Schur-complement extension adjusts each local score for sibling-cluster dependence with quantitative variance reduction bounds (Section 7).
- (vi) **Statistical validity.** Under i.i.d. sampling with fixed dimension and dendrogram, the sample RA-HRP portfolio is consistent and asymptotically normal at the $\sqrt{T}$ rate, with explicit Jacobian structure (Theorem 8.3).
- (vii) **Dendrogram stability.** Empirical diagnostics show moderate tree stability (70.8% mean merge preservation, 0.987 cophenetic correlation) with episodes of instability,

establishing that the fixed-tree theory is a useful approximation but not a complete description (Section 11.5).

- (viii) **Comparative robustness.** The fixed-tree RA-HRP map is locally Lipschitz with explicit constant on compact neighbourhoods away from vanishing cluster variances, whereas the tangency-portfolio Jacobian diverges with $\lambda_{\min}(\Sigma)^{-1}$ (Propositions 8.6–9.1).
- (ix) **Spectral benchmarks with projection error bound.** Section 10 clarifies that Schur EW and Schur RA are simplex-projected spectral directions, whereas Schur MV is best interpreted empirically as a regularised spectral GMV-like comparator. Theorem 10.12 bounds the distance between the long-only projected tangency portfolio and the unconstrained tangency portfolio in terms of aggregate short mass, quantifying when the long-only constraint is binding. In the reported backtest, RA-HRP achieves a Sharpe ratio of 1.349 and a 7.76 percentage point annual excess return over traditional HRP, while the capped market-cap proxy achieves a higher Sharpe (1.571), underscoring the regime-dependence of the results.

The main contribution is therefore a mathematically complete fixed-tree theory for RA-HRP—including quantitative bounds, structural decompositions, and asymptotic guarantees—together with evidence that, in a favourable but nontrivial US equity sample, modest local return tilts can improve upon pure HRP without reverting to unstable global mean–variance optimisation. The open questions are: (i) extending the asymptotic theory to estimated dendrograms, (ii) establishing whether RA-HRP’s edge survives broader asset universes and unfavourable regimes, and (iii) developing alternative return estimators that are more robust to structural breaks.

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